IT8313 Wireless Communication Project PBL Project (5G Infrastructure Deployment)

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Submitted to: **Syed Imran Patel**

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Abstract

This project presents the deployment of 5G wireless technology, the latest advancement in mobile networks operating at ultra-fast speeds with reliable connectivity.

The objectives include designing a scalable 5G infrastructure for Bahrain, optimizing coverage for urban and suburban areas, and evaluating performance through simulations.

The project infrastructure involves detailed RF planning, cell classification, and OMNeT++-based simulations to model network behaviour under varying traffic loads.

Results demonstrate throughput, reduced latency, and efficient resource allocation in 5G design.

In conclusion, this project designed a 5G network plan for Bahrain, focusing on better coverage and faster speeds and provides steps for making this happen.

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Introduction

The goal of this project is to design and implement a 5G wireless network for OKAKNet Telecom in Bahrain. It is all geared toward making the country use fast, technically better, and maximally well-covered. This will offer huge benefits to sectors such as healthcare, education, government services, and transportation and individuals.

The project has two main sets of objectives. First, there are general objectives aimed at business growth for OKAKNet by extending the 5G services to areas that currently lack good connectivity on a good quality basis to accommodate different type of users. Second, the technical objectives involve planning the network, designing the layout, testing how it handles heavy traffic, and finally running simulations to evaluate performance in both busy cities and quieter suburban areas. The project will also include recommendations for future improvements to keep the network up to date.

This project has both a theory and practical implementation to create realistic simulations based on OMNeT++. The study results will help OKAKNet in building a network for the present needs of Bahrain and its further development.

Description of the report

This study serves as a 5G infrastructure deployment analysis and practical approach for Bahrain. The goal is to improve connectivity across the country by increasing speed and available bandwidth as well as reliability and coverage. In addition, serving all sectors including healthcare, education, government, and mass transport.

The first chapter is a discussion on the various wireless technologies, including WLAN, Satellite Networks, and Wireless Cellular Networks. It then goes into some detail on the various generations of cellular networks, such as 2G, 3G, 4G, and 5G. A comparison between 4G and 5G has been made to highlight improvements, and reasons are given as to why 5G is the favoured technology.

The second chapter focuses on the network planning and design for each governorate in Bahrain. Each area is divided into zones such as residential, industrial, and commercial. The population density for each governorate is studied, and areas are classified as urban or suburban. Cell classification is done based on these areas, and path loss calculations are made to estimate the coverage of UMi and UMa cells. The number of cells required for each area is then calculated and distributed accordingly. After that, the cell peak throughput and the number of users per cell are estimated based on the available resource blocks.

The third chapter shows the network design implementation using OMNeT++ with graphs and explanations for each scenario.

Wireless Technology Introduction

Wireless technologies

Wireless technology: It is a technology that allows devices to communicate over the air interface “or in another word communicate without cables or wires” if the device support that technology. There are many types of wireless technology some of them are:

- WLAN (Wireless Local Area Networks)
- Satellite Networks
- Wireless Cellular Networks

WLAN (Wireless Local Area Network)

WLAN (Wireless Local Area Network): Is a wireless computer network that allow two or more devices using the air interface to form local area network to communicate with each other within a limit range of area such as: home, work, office building or campus. WLAN is based on the IEEE 802.11 standards, and it consider one of the most used computer networks in the world these days also WLAN uses multiple versions of WI-FI (Wireless Fidelity) here is a brief information about the last five version of WI-FI:

WI-FI Version	IEEE Standards	Release Year	Theoretical Max Speed	Frequency Band
WI-FI 5	802.11ac	2014	3.5 Gbps	5 GHz
WI-FI 6	802.11ax	2019	9.6 Gbps	2.4, 5 GHz
WI-FI 6E	802.11ax	2021	9.6 Gbps	6 GHz
WI-FI 7	802.11be	2024	46 Gbps	2.4, 5 and 6 GHz

Table 1 WLAN Specification

Advantages of WLAN:

- No cables
- Easy mobility
- Easier setup in term of connectivity

Disadvantages of WLAN:

- Limit range of coverage
- Slower its counterpart the wired connection
- Security risks

Satellite networks

A Satellite networks: are multiple orbiting satellites working together to form a network to provide communication and other service over long distances especially in remote areas where ground-based infrastructure (like cellular towers or fibre optics) is unavailable. These networks rely on very complex and sophisticated technology to facilitate information exchange between satellites and users on Earth. The satellite networks need several components to work:

- The satellite
- A ground station
- Microwave

Types of Satellite Networks:

Orbit type	Altitude	Coverage	Latency
Low earth orbit (LEO)	160 – 2000 km	Small area	20 – 50 ms
Medium earth orbit (MEO)	2000 – 35786 km	regional	100 – 150 ms
Geostationary Orbit (GEO)	35786 km and above	global	500 – 700 ms

Table 2 Satellite Networks Types

Advantages of satellite networks:

- Geographical area coverage
- Scalability
- 24/7 continuous operation

Disadvantages:

- Require high cost
- Weather & Signal Interference
- High Latency

Wireless Cellular Networks

Wireless Cellular Networks: Also known as mobile network is a telecommunication network where the link to and from the nodes is wireless and the network is distributed over land areas called cells, each cell aids by at least one base transceiver station (BTS). These BTS units provide their own cells with network coverage, allowing the transmission of voice, data, and other content via radio waves. Multiple BTS units are managed and coordinated by a Base Station Controller (BSC).

Key components of a cellular network:

- IoT Devices are sensors, actuators and smart gadgets that are designed to gather data, control physical system within the ecosystem.
- IoT Gateway: devices that acts as a bridge between IoT devices and the border network infrastructure to provide all-in-one connection and data exchange with the other side of the network.
- Base Transceiver Station (BTS): Towers that equipped with antennas and transceivers that simplify the communication.
- Base Station Controller (BSC): the BSC is the second most important components in the mobile network ecosystem and its main purpose is to manages multiple base stations across a single cellular network.
- Mobile Switching Center (MSC): is a central switching unit that connects the cellular network to other networks.
- Home Location Register (HLR): is a database used in cellular network to store the subscriber information.
- Visitor Location Register (VLR): it is a temporary database that stores the subscriber information if they roam outside their home network coverage area.
- Authentication Center (AuC): main purpose of it to authenticate and verify the identity of the subscribers.
- Gateway Mobile Switching Center (GMSC): is an interface between the external network and the cellular network.

Discuss cellular network types

Cellular networks are classified into generations (1G to 5G), each offering advancements in speed, connectivity, and functionality. **1G** introduced analogue voice calls, while **2G** brought digital encryption and SMS. **3G** enabled mobile internet, and **4G LTE** revolutionized broadband with high-speed data for streaming and VoIP. The latest, **5G**, delivers ultra-fast speeds, ultra-low latency, and supports IoT and smart technologies.

2G networks

2G referred to as the second generation of cellular network, 2G cellular telecom networks were launched on the standards of the Global Communication Systems (GSM) in 1991 worldwide and Code Division Multiple Access IS-95 (CDMA) in the united state of America.

2G brings 3 primary benefits over their predecessor (1G ,1.5G) like phone conversation are digitally encrypted, 2G system are more efficient also, 2G introduced data services for mobile begging with the Short Message Service (SMS) plain text messages. 2G technologies allows numerous mobile phone network to provide the services such as picture messages, text messages and multimedia message services (MMS).

2G telecom has some enhanced versions the 2.5G General Packet Radio Service (GRRS) also, referred to it as “Second-and-a-half generation” and 2.75G Enhanced Data Rates for GSM Evolution (EDGE).

3G networks

3G referred to as the third generation of cellular network was launched around the begging of the 2000s and provides a massive advancement over the last generation specifically in terms of data transfer speeds and mobile internet.

The 3G network were developed on two different standards UTMS by 3GPP succeeding the GSM and CDMA2000 developed by Qualcomm succeeding cdmaOne. Both were based on the IMT-2000 specifications established by the International Telecommunication Union (ITU).

Later 3G have some enhancement releases, often referred to them as 3.5G and 3.75G.

4G networks

4G refers to the fourth generation of mobile telecommunication network published in the late 2000s and early 2010s. Compared to the previous generation 3G, 4G is designed to support all IP communication and services and eradicates the circuit switching, having higher data bandwidth compared with 3G. allowing a diversity of data- intensive application and services.

The original deployed technologies of the 4G were Long Time Evolution (LTE) created by 3GPP group and Mobile Worldwide Interoperability for Microwave Access (Mobile WiMAX), Based on IEEE specification. This specification provides significant enhancements over prior Technology 2G and 3G.

5G Networks

5G network consider the most major phase in mobile communication beyond 4G standards and have been developed by mobile operator’s worldwide sine since 2019. 5G networks offer not only higher downloads speeds with a peak speed of 10 gigabits per second (Gbit/s) with lower latency.

3GPP has been developed a global unified standards and they called it 5G New Radio (5G NR) based on the specification provided by the International Telecommunication Union (ITU) under the IMT-2000 requirements.

The increased bandwidth that comes with the 5G from previous generation which is 4G brings the capability of connecting more devices simultaneously with the improvement of the quality of the cellular data when being crowded. These upgraded versions of the features make the 5G particularly suited for applications requiring real-time data exchange.

5G deployment bring some challenges when it such as significant investment, spectrum allocation, security risks, and concerns about energy efficiency and environmental impact accompanying with the use of higher frequency bands.

Research and comparison of 5G with technologies

5G technology

marks a substantial advancement in wireless communication with rapid speed, less latency, higher bandwidth, and the connectivity increased compared to other generations before like 4G. It makes use of three primary frequency bands: high-band millimeter wave for extremely fast speeds in crowded places as well as short distances. Mid-band for balanced performance which can cover large area as well as low band for broad coverage which plays a key role in expanding 5G coverage by providing advantageous radio wave propagation properties. Its flexible architecture ensures efficient spectrum use while supporting diverse industry needs.

Band Type	Frequency Range	Speed	Area
Low	600 MHz – 1 GHz	Lower speed: ~100 Mbps	Less dense area
Mid	1GHz – 6 GHz	Balanced speed: 300 Mbps – 1 Gbps	Urban & suburban
High	24 GHz – 100 GHz	Ultra-high speed < 1 Gbps	Crowded

Table 3 5G Technologies

5G vs 4G (Comparison)

Boundary		5G	4G LTE
Frequency		**Low Band:** (< 1 GHz), **Mid Band:** Sub-6 GHz (1–6 GHz), **High Band:** mmWave (24–100 GHz)	700 MHz – 3.5 GHz
Bandwidth spectrum		*Up to 100 to 800 MHz (mmWave), 100 MHz (Sub-6 GHz)*	Up to 20 MHz
Multiplexing		**OFDMA** (Orthogonal Frequency Division Multiplexing Access)	**OFDM** (Orthogonal Frequency Division Multiplexing)
Modulations		QPSK, (16, 64, 256, 1024 QAM)	16, 64, 256 QAM
Data Rate Support	UL (Uplink)	**50** Mbps to **10** Gbps	**2** Mbps to **50** Mbps
	DL (Downlink)	**100** Mbps to **20** Gbps	**5** Mbps to **300** Mbps
MIMO		**Massive MIMO** (massive multiple-input multiple-output)	**MIMO** (Multiple-Input Multiple-Output)
Core Network (All IP/ Hybrid)		**5GC** (5G Core), Cloud Native	**EPC** (Evolved Packet Core)
Standard Organization		3GPP (Release 15 onwards)	3GPP (Release 8–14)

Table 4 Comparison 4G vs 5G

Justification of 5G

Since OKAKNet wants to improve wireless services for governments, hospitals, schools, and busy cities, 5G is the best choice to ensure the best speed and lowest latency. For example, schools can run online classes without lag, and more people in cities can use the internet at the same time. Also, 5G is better for urban areas to maintain strong signals with many users and for suburbs to stay connected without dead zones. To conclude, 5G is faster, more reliable, and more future-ready than 4G and older technologies.

Network Planning and Designing

Governorates of Bahrain:

Capital Governorate:
Capital population are around 536,933 and an area of 79 Km^2. It contains multiple areas including residential areas like Al Eker, Sanabis, Sanad and Jurdab, Industrial areas like Industrial Sitra, Minaa Salman and commercial areas like Commercial Manama and Seef.

Muharraq Governorate:
Muharraq governorate area is 74 Km^2 and citizens around 283,372. It includes Muharraq City, Arad, Al Hidd, Qalali and Diyar Al Muharraq as residential areas. Industrial areas such as Khalifa bin Salman Port and Al Hidd industrial as well as airport and we calculate airport area with Muharraq City.

Northern Governorate:
The **Northern Governorate** covers an area of 146 km^2 and has a population of around 415,779 people. It includes many residential areas such as Karranah, Budaiya, Dar Kulaib, and Hamad Town, as well as open spaces like Um Al Nassan.

Southern Governorate:
The southern governorate covers an area of 489 km^2 with a population 321,098. The governorate is divided into three areas residential area like Awali, Isa town, Riffa, Al Mazrowiah and industrial area like southern industrial area and educational area like University of Bahrain and the educational area in Isa Town.

Population and area for each Governorates:

The following report presents the population of the governorate of Bahrain. The capital Governorate has the highest population with approximately 536,933 residents. The Northern Governorate follows as the second highest with approximately 415,779, while the Southern Governorate has 321,098 approximately. Finally, the Muharraq governorate has the least population with 283,372 residents approximately.

Governorates	Population (Sum)	Area (km^2)
Capital	536,933	79
Muharraq	283,372	74
Northern	415,779	146
Southern	321,098	489

Table 5 Governorate Population and Area

Density for each Governorates:

To calculate the density for each governorate we will use the following formula:

Density formula:

$$\boldsymbol{Density = \frac{Total\ Population}{Area\ (km^2)}}$$

All number here are rounded to the nearest number:

Capital Density:

$$\boldsymbol{Density} = \frac{\mathbf{536,933}}{\mathbf{79}} = \mathbf{6797}$$

Muharraq Density:

$$\boldsymbol{Density} = \frac{283{,}372}{\mathbf{74}} = \mathbf{3829}$$

Northen Density:

$$\boldsymbol{Density} = \frac{415{,}779}{\mathbf{146}} = \mathbf{2848}$$

Southern Density:

$$\boldsymbol{Density} = \frac{321{,}098}{\mathbf{489}} = \mathbf{657}$$

Cell classification for each governorate:

Governorate	Area Type	Area	Population	Density
Capital	Dense urban	79 Km^2	536,933	6797
Muharraq	Urban	74 Km^2	283,372	3829
Northern	Urban/Suburban	146 Km^2	415,779	2848
Southern	Open space (Mostly Rural)	489 Km^2	321,098	657

Table 6 Cell classification for each governorate

Capital Governorate	Area (Km^2)	Area Types
Residential Areas	**Total: 9.35**	
Al Eker	2.68	Suburban
Sanabis	1.62	Urban
Sanad	3.85	Urban
Jurdab	1.2	Urban
Industrial Areas	**Total: 17**	
Industrial Sitra	15	Suburban
Minaa Salman	2	Suburban
Commercial Areas	**Total: 17.78**	
Commercial Manama	14.15	Suburban
Seef	3.63	Urban
	Total Areas: 44.13	

Table 7 Capital Governorate Areas

Muharraq Governorate	Area (Km^2)	Area Types
Residential Areas	**Total: 33.56**	
Muharraq City	4.88	Urban
Arad & Al Hidd	12.7	Urban
Qalali	3.01	Urban
Diyar Al Muharraq	12.97	Suburban
Industrial Areas	**Total: 17.81**	
Al Hidd Industrial	3.24	Suburban
Khalifa Bin Salman Port	14.57	Suburban
Airport	**Total: 5.94**	
Bahrain international Airport	5.94	Urban
	Total Area: 49.49	

Table 8 Muharraq Governorate Areas

Northern Governorate	Area (Km^2)	Area Types
Residential Areas	**Total: 24.35**	
Dar Kulaib	2.06	Suburban
Karzakan	3.09	Suburban
Karranah	2.85	Urban
Hamad Town	14.02	Urban
Budaiya	2.33	Urban
Open Spaces	**Total: 21.71**	
Um Al Nassan	21.71	Suburban
	Total Area: 46.06	

Table 9 Northern Governorate Areas

Southern Governorate	Area (Km^2)	Area Types
Residential Areas	**Total: 44.42**	
Isa Town	5.02	Urban
Al Mazrowiah	4.23	Urban
Awali	3.43	Suburban
Riffa	31.74	Urban
Educational Areas	**Total: 4.35**	
University of Bahrain	2.74	Suburban
Educational Area	1.61	Urban
Industrial areas	**Total: 21.39**	
Industrial Area	21.39	Suburban
	Total Area: 70.2	

Table 10 Southern Governorate Areas

Coverage of Areas:

When calculating the number of cells for each governorate, we need to use the path loss equation. Choosing the power received (Pr) value depends on a lot of the governorate environment such as urban and suburban. Pr value should be above -100 dBm for 5G technology to ensure signal can be detected by all nearby supported devices. Above -90 dBm or value is considering an acceptable value to provide a good signal and range. After obtaining the distance from the equation we need to get the one cell radius coverage area by using the formula:

$$\boldsymbol{One\ cell\ coverage\ area} = \boldsymbol{\pi r^2}$$

Path loss modelling

Calculation was performed in excel using path loss model to calculate the area for each category in each governorate. The categories are divided into urban and suburban, where Macro cells used for suburban areas and Micro cells for urban areas.

UMa (Urban Macro Cells) one cell coverage:

Here the input of the calculations:

Refer to the Appendix 1 - UMa Parameters

Power transmitted	50	Watt	in dBm	46.9897
Frequency	3.5	GHz		
BS Height	35	Meter		
UE Height	2	Meter		
MIMO	64	No. Antenna		

Table 11 UMa Cells Parameters

To calculate the one cell coverage for the UMa, we need to calculate the path loss and power received

Start with the path loss using the below formula:

$$Path\ loss = 13.54 + 39.08 \log_{10}(D) + 20\log_{10}(F) - 0.6 \log_{10}(UE - 1.5)$$

Substitute the values into the formula:

$$Path\ loss = 13.54 + 39.08 \log_{10}(680) + 20\log_{10}(3.5) - 0.6 \log_{10}(2 - 1.5) = 135.2964272$$

Then calculate the power received using this formula:

$$Power\ reveived = (Pt + 10 \log_{10}(MIMO)) - path\ loss$$

Next Substitute the values into the formula:

$$Power\ reveived = (46.9897 + 10 \log_{10}(64)) - 135.2964272 = -70.24492745$$

Refer to the Appendix 2 – UMa Path Loss and Power received

620	133.7286461	-68.67714637
630	134.0002076	-68.94870782
640	134.2674923	-69.21599253
650	134.5306329	-69.47913312
660	134.7897559	-69.73825615
670	135.0449822	-69.99348243
680	135.2964272	-70.24492745
690	135.5442014	-70.49270161
700	135.7884103	-70.73691055
710	136.0291552	-70.97765541

Table 12 UMa Path Loss and Power Recevied

The result shows that at 680 meter the Pr value is -70.24492745 will fails under the acceptable value for the governorate. To convert the 680 meters to a radius you should use the one cell coverage formula.

$$\boldsymbol{One\ cell\ coverage = \pi r^2}$$

Since we chose --70.24492745 as the power received at 680 meters. We can substitute:

$$\boldsymbol{One\ cell\ coverage = \pi * 680 * 680 = 1452672.443\ m^2}$$

Now, divide the answer by 1,000,000 to convert the m^2 into km^2:

$$\frac{\mathbf{1452672.443}}{1.000{,}000} = 1.45\ km^2$$

So, now the total one cell coverage is **1.45 km²**

From the calculations above, we notice that the received power declines as we move farther from the cell. After evaluating different coverage distances, we determined that 680m is the optimal choice, as it ensures the received power remains above -90 maintaining a strong signal while minimizing interference.

Additionally, the 680m radius reduces the number of required cells compared to smaller coverage distances, leading to cost savings in infrastructure deployment.

Choosing a 680m radius provides the best balance between signal quality, coverage efficiency, and cost-effectiveness. A smaller radius would require more cells, increasing deployment and maintenance expenses without significant performance benefits. Therefore, 680m is the optimal radius for this network design.

UMi (Urban Micro Cells) one cell coverage:

The input of the calculations:

Refer to the Appendix 4 – UMi Parameters

Pt	100	Watt	in dBm	50
Frequency	24	GHz		
BS Height	35	Meter		
UE Height	2	Meter		
MIMO	128	number of antennas		

Table 13 UMi Cells Paramaters

To calculate the one cell coverage for the UMi, we need to calculate the path loss and power received.

Start with the path loss using the below formula:

$$Path\ loss = 13.54 + 39.08\log_{10}(D) + 20\log_{10}(F) - 0.6\log_{10}(UE - 1.5)$$

Substitute the values into the formula:

$$Path\ loss = 13.54 + 39.08\log_{10}(460) + 20\log_{10}(24) - 0.6\log_{10}(2 - 1.5) = 145.3854189$$

Then calculate the power received using this formula:

$$Power\ reveived = (Pt + 10\log_{10}(MIMO)) - path\ loss$$

Next Substitute the values into the formula:

$$Power\ reveived = (50 + 10\log_{10}(128)) - 145.3854189 = -74.3133192$$

Refer to the Appendix 5 – UMi Path Loss and Power received

420	143.8414251	-72.7693254
430	144.2407901	-73.16869038
440	144.6309734	-73.55887373
450	145.0123879	-73.94028817
460	145.3854189	-74.3133192
470	145.7504271	-74.67832742
480	146.1077504	-75.03565069
490	146.4577056	-75.38560594
500	146.8005906	-75.7284909
420	143.8414251	-72.7693254

Table 14 UMi Path Loss and Power Recevied

The result shows that at 460 meter the Pr value is -74.3133192 will fails under the acceptable value for the governorate. To convert the 460 meters to a radius you should use the one cell coverage formula:

$$\boldsymbol{One\ cell\ coverage = \pi r^2}$$

Since we chose -74.3133192 as the power received at 460 meters. We can substitute:

$$One\ cell\ coverage = \pi * 460 * 460 = 664761.005\ m^2$$

Now, divide the answer by 1,000,000 to convert the m^2 into km^2:

$$\frac{664761.005}{1.000,000} = 0.66\text{km}^2$$

So, now the total one cell coverage is **0.66 km²**

Unlike macro cells, micro cells are designed for smaller, high-density areas, requiring a reduced coverage range. After the calculations, we determined that a 460m radius is the optimal choice for micro cells, as it maintains the received power above -90, ensuring strong signal quality while minimizing interference in urban or crowded environments.

Additionally, the 460m radius reduces the number of required cells compared to a smaller radius, leading to cost-efficient deployment.

Choosing a 460m radius hit the best balance between signal reliability, network capacity, and cost-efficiency. A smaller radius significantly increases the number of cells, raising deployment and maintenance costs without proportional benefits. Therefore, 460m is the recommended radius for micro cell deployment in high-traffic zones.

Cells per Area

To calculate the number of cells for each City we need to obtain the area if the city then divide it by the one cell coverage area:

$$\textbf{Number of cell per city} = \frac{\textbf{Area}(\textbf{km}^2)}{\textbf{Total one cell coverage}\ (\textbf{km}^2)}$$

Total number for each Governorate areas:

Capital Governorate:

Capital Governorate	Total Cells	Area type	Cell Type
Residential Areas	**12 Cells**		
Sanabis	2	Urban	UMi
Sanad	6	Urban	UMi
Jurdab	2	Urban	UMi
Al Eker	2	Suburban	UMa
Commercial Areas	**15 Cells**		
Commercial Manama	10	Suburban	UMa
Seef	5	Urban	UMi
Industrial Areas	**11 Cells**		
Industrial Sitra	10	Suburban	UMa
Minaa Salman	1	Suburban	UMa

Table 15 Capital Governorate Total Cells

Muharraq Governorate:

Muharraq Governorate	Total Cells	Area type	Cell Type
Residential Areas	**40 Cells**		
Diyar Al Muharraq	9	Suburban	UMa
Muharraq City	7	Urban	UMi
Arad & Al Hidd	19	Urban	UMi
Qalali	5	Urban	UMi
Airport	**9 Cells**		
Bahrain National Airport	9	Urban	UMi
Industrial Areas	**12 Cells**		
Al Hidd Industrial	2	Suburban	UMa
Khalifa Bin Salman Port	10	Suburban	UMa

Table 16 Muharraq Governorate Total Cells

Northern Governorate:

Northern Governorate	Total Cells	Area type	Cell Type
Residential Areas	**32 Cells**		
Dar Kulaib	1	Suburban	UMa
Karzakan	2	Suburban	UMa
Karranah	4	Urban	UMi
Hamad Town	21	Urban	UMi
Budaiya	4	Urban	UMi
Open spaces	**15 Cells**		
Um Al Nasan	15	Suburban	UMa

Table 17 Northern Governorate Total Cells

Southern Governorate:

Southern Governorate	Total Cells	Area type	Cell Type
Residential Areas	**64 Cells**		
Isa Town	8	Urban	UMi
Al Mazrowiah	6	Urban	UMi
Awali	2	Suburban	UMa
Riffa	48	Urban	UMi
Industrial areas	**15 Cells**		
Industrial Area	15	Suburban	UMa
Educational Areas	**4 Cells**		
Educational Area	2	Urban	UMi
University of Bahrain	2	Suburban	UMa

Table 18 Southern Governorate Total Cells

5G Network Topology:

After getting the total cells for each city. Here is the distribution of each cell.

Capital Governorate:

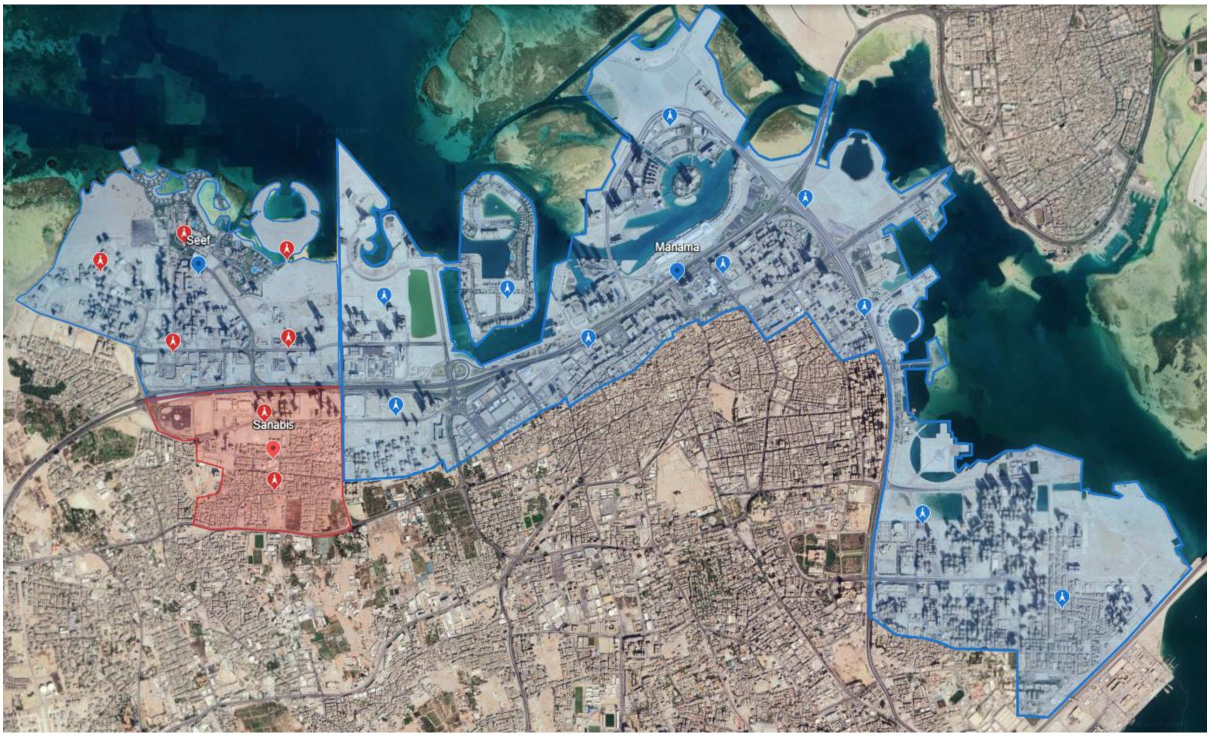


Figure 1 Capital Governorate 5G Topology #1

Figure 2 Capital Governorate 5G Topology #2

Muharraq Governorate:

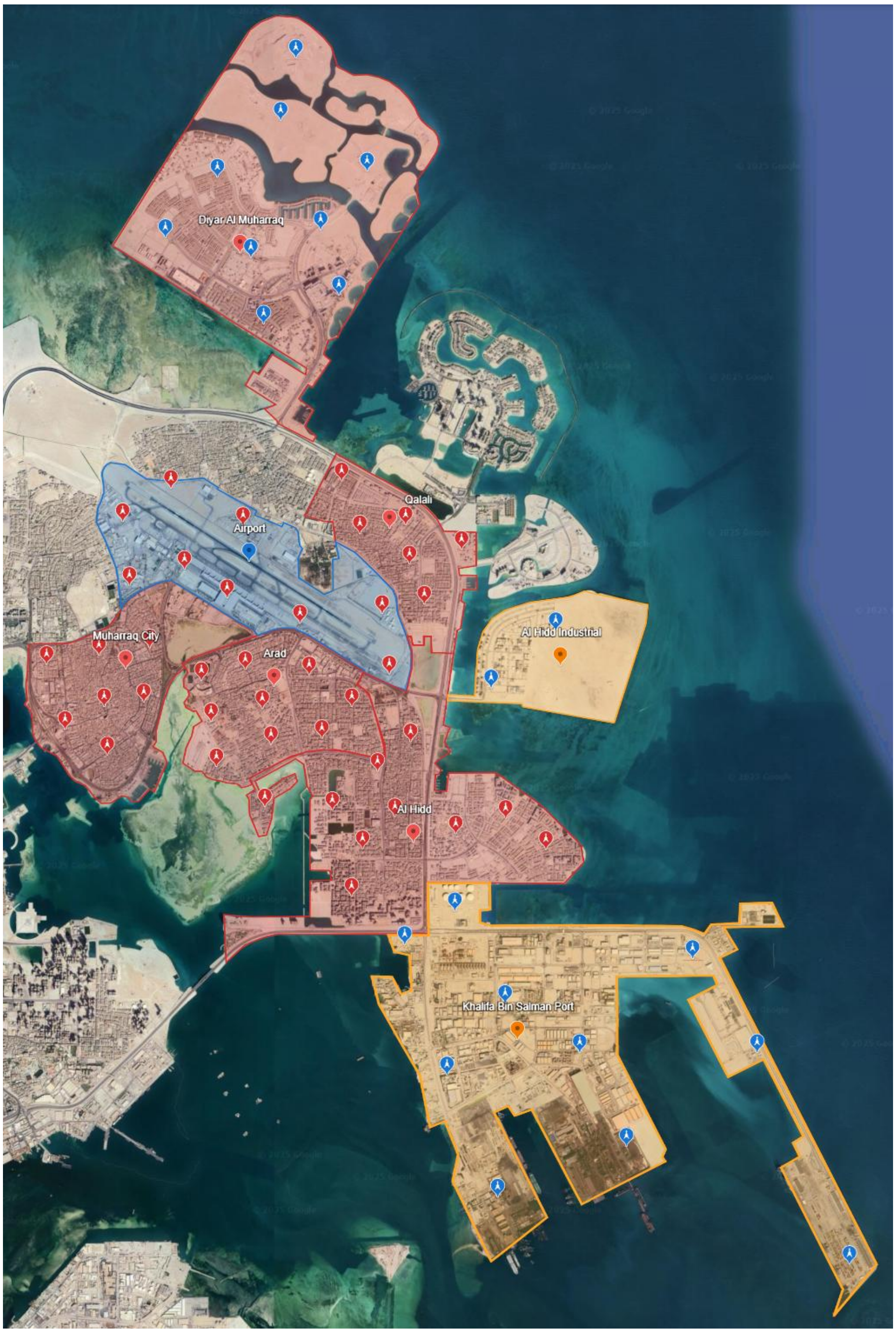


Figure 3 Muharraq Governorate 5G Topology

Northern Governate:

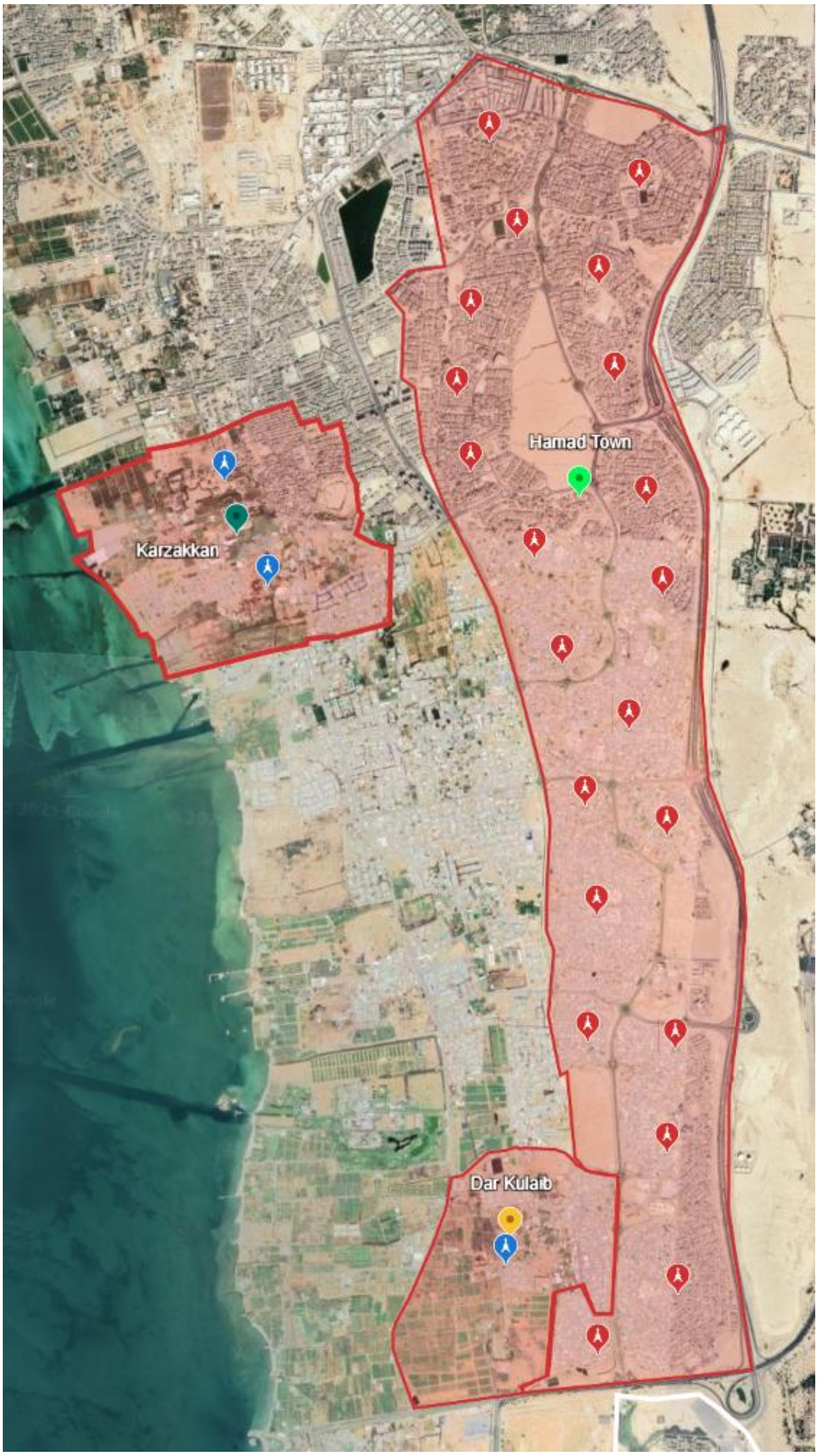


Figure 4 Northern Governorate 5G Topology #1

Figure 5 Northern Governorate 5G Topology #2

Figure 6 Northern Governorate 5G Topology #3

Southern Governorate:

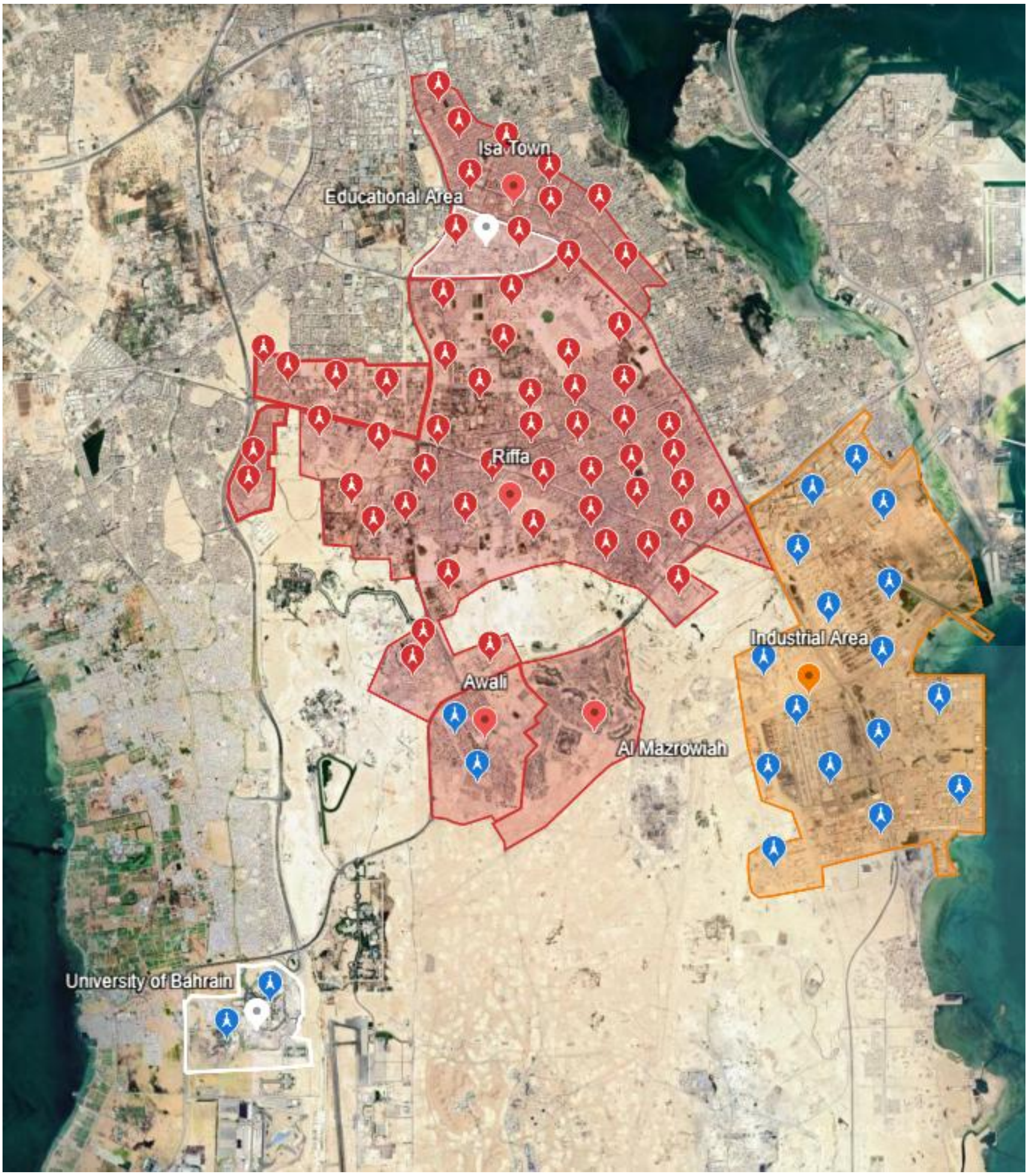


Figure 7 Southern Governorate 5G Topology

Cell Peak Throughput

Cell peak throughput is calculated by using Number of Resource Block (NBR), Number of layers and Throughput per RB.

$$Cell\ peak\ throughput = \frac{Throughput\ per\ RB * Number\ of\ RB * Number\ of\ Layer}{10^6}$$

However, we need to determine the bandwidth and SCS because Number of Resource Block are depended on both of them. Based on the 3GPP specification FR1 and FR2 have different SCS ranges. FR1 (1-6 GHz) in Figure 8 FR1, and FR2 (24, 26, 28, 39 GHz), in Figure 9 FR2

$$Number\ of\ resource\ blocks = \frac{Bandwidth - Guard\ bandwidth}{Subcarrier\ spacing * 12}$$

Table 5.3.2-1: *Transmission bandwidth configuration* N_{RB} for FR1

SCS (kHz)	3 MHz	5 MHz	10 MHz	15 MHz	20 MHz	25 MHz	30 MHz	35 MHz	40 MHz	45 MHz	50 MHz	60 MHz	70 MHz	80 MHz	90 MHz	100 MHz
	N_{RB}	N_{RB}	N_{RB}	N_{RB}	N_{RB}	N_{RB}	N_{RB}	N_{RB}	N_{RB}	N_{RB}	N_{RB}	N_{RB}	N_{RB}	N_{RB}	N_{RB}	N_{RB}
15	15	25	52	79	106	133	160	188	216	242	270	N/A	N/A	N/A	N/A	N/A
30	N/A	11	24	38	51	65	78	92	106	119	133	162	189	217	245	273
60	N/A	N/A	11	18	24	31	38	44	51	58	65	79	93	107	121	135

Figure 8 FR1

Table 5.3.2-2: *Transmission bandwidth configuration* N_{RB} for FR2-1

SCS (kHz)	50 MHz	100 MHz	200 MHz	400 MHz
	N_{RB}	N_{RB}	N_{RB}	N_{RB}
60	66	132	264	N/A
120	32	66	132	264

Figure 9 FR2

Number of throughputs per resource block (RB) and Number of layers (NL). Both of them are considered fixed values in 5G. We will only change the bandwidth and Subcarrier spacing (SCS) anything else we will the same which are the standard values.

Since we are following 3GPP based on their specification release 19. The choice of the SCS depends entirely on the selected bandwidth. In our deployment we for the 5G we used FR1 at 3.5 GHz which slides under the n78 band and FR2 at 24 GHz which slides under the n258 band. The Subcarrier spacing will be determined by selecting bandwidth (BW) first. According to the Figure 8 FR1, for example if we chose 100 MHz as the bandwidth, there are 3 SCS options available to choose from: 15 KHz, 30 KHz and 60 KHz. From Figure 9 FR2 the available SCS are 60 KHz and 120 KHz.

The Subcarrier Spacing is related to the latency. The Higher the Subcarrier Spacing value, the lower the latency will be on the system.

Chosen bandwidth is depended on the average data rate. If there is an area that have higher data rate demand it is better to increase the bandwidth so it can accommodate more and more users at once.

Spectral efficiency (SE) parameter in our deployment will be 256 QAM which the modulation order value will be 8 bits. Coding rate has a range starting from 0.7 until 0.9 which depends on the reliability. For our installation of 5G network our coding rate 0.9.

$$\text{Spectral efficiency (SE)} = \text{Modulation Order} * \text{ Coding rate}$$

$$Modulation\ Order = \log_2(256) = 8$$

$$\text{Spectral efficiency (SE)} = 8 * \ 0.9 = 7.2$$

To estimate the bandwidth and the SCS and the user per cell for each category. We need to know which service are being used in every category to obtain the average user per demand.

$$User\ per\ cell = \frac{Peak\ throughput * (100\% - Management\ overhead\%)}{Average\ demand}$$

Residential areas use the most common services like Video streaming, Web browsing, Video calling, social media and Online gaming. Online gaming requires up to 20 Mbps, 4K Video streaming uses up to 25 Mbps, Web browsing demands 5 Mbps, social media requires up to 10 Mbps and Video calling requires up to 5 Mbps.

Service	Bandwidth Required	Average demand (Mbps)
Video streaming	25 Mbps	13 Mbps
Web browsing	5 Mbps	
Video calling	5 Mbps	
Social media	10 Mbps	
Online gaming	20 Mbps	

Table 19 Residential Area Demand

Since Residential areas have both Micro and Macro.

Micro:

To meet the high demand of the Residential areas, the selected bandwidth must be high so, according to the Figure 9 FR2the selected bandwidth for the Micro cells will be 200 Mbps so it can support as much user.

Since we are using high demand service like online gaming the low latency is needed so, the 60 KHz is the most suitable and balanced between throughput and latency.

Bandwidth (MHz)	Subcarrier spacing (KHz)
200 MHz	60 KHz

Table 20 Residential Area Bandwidth/SCS for Micro

Refer to the Appendix 7 - Residential Area Micro Throughput Calculations

Calculating the Number of resource block (NRB):

$$\text{number of resource block} = \frac{(200 * 10^6 - 1.68 * 10^6)}{60 * 10^3 * 12} = 275$$

Calculating Peak throughput:

$$\boldsymbol{Cell\ peak\ throughput = \frac{(2.4192 * 10^6) * 275 * 4}{10^9} = 2.66112\ Gbps}$$

Calculating User per cell:

$$\boldsymbol{User\ per\ cell} = \frac{(\mathbf{2.66112} * \mathbf{10^3}) * (\mathbf{100\%} - \mathbf{20\%})}{\mathbf{13}} = \mathbf{163.761}$$

Macro:

Since the suburban areas are consuming approximately the same bandwidth for services that are used in the urban areas. Also, the standard bandwidth for FR1 macro cells is 100 MHz for less dense areas.

According to Figure 8 FR1 the standard SCS for FR1 ranges is 30 KHz to reach the average latency requirement for the service to ensure balancing.

Bandwidth (MHz)	Subcarrier spacing (KHz)
100 MHz	30 KHz

Table 21 Residential Area Bandwidth/SCS for Macro

Refer to the Appendix 8 - Residential Area Macro Throughput Calculations

Calculating the Number of resource block (NRB):

$$\text{number of resource block} = \frac{(100 * 10^6 - 1.68 * 10^6)}{30 * 10^3 * 12} = 273$$

Calculating Peak throughput:

$$\boldsymbol{Cell\ peak\ throughput} = \frac{(\mathbf{2.4192} * \mathbf{10^6}) * \mathbf{273} * \mathbf{4}}{\mathbf{10^9}} = \mathbf{2.6417664}\ \boldsymbol{Gbps}$$

Calculating User per cell:

$$\boldsymbol{User\ per\ cell} = \frac{(\mathbf{2.6417664} * \mathbf{10^3}) * (\mathbf{100\%} - \mathbf{20\%})}{\mathbf{13}} = \mathbf{162.570}$$

Commercial areas use services like Web browsing, Mobile payments, social media and cloud-based business applications. Web browsing demands 5 Mbps, mobile payments up to 1 Mbps, social media 10 Mbps and cloud-based business applications consumes around 15 Mbps.

Service	Bandwidth Required	Average demand (Mbps)
Web browsing	5 Mbps	7.75
Mobile payments	1 Mbps	
social media	10 Mbps	
cloud-based business applications	15 Mbps	

Table 22 Commercial Area Demand

Commercial Areas used both UMa cells and UMi cells.

Micro:

Since commercial area uses unique service that needs to be run even under load. The chosen bandwidth will be 200 MHz with a subcarrier spacing value 60 KHz.

Bandwidth (MHz)	Subcarrier spacing (KHz)
200 MHz	60 KHz

Table 23 Commercial Area Bandwidth/SCS for Micro

Refer to the Appendix 9 - Commercial Area Micro Throughput Calculations

Calculating the Number of resource block (NRB):

$$\text{number of resource block} = \frac{(200 * 10^6 - 1.68 * 10^6)}{60 * 10^3 * 12} = 275$$

Calculating Peak throughput:

$$\boldsymbol{Cell\ peak\ throughput = \frac{(2.4192 * 10^6) * 275 * 4}{10^9} = 2.66112\ Gbps}$$

Calculating User per cell:

$$\boldsymbol{User\ per\ cell = \frac{(2.66112 * 10^3) * (100\% - 20\%)}{7.75} = 274.696}$$

Macro:

Since the suburban areas are consuming approximately the same bandwidth for services that are used in the urban areas. Also, the standard bandwidth for FR1 macro cells is 100 MHz for less dense areas.

According to Figure 8 FR1 the standard SCS for FR1 ranges is 30 KHz to reach the average latency requirement for the service to ensure balancing.

Bandwidth (MHz)	Subcarrier spacing (KHz)
100 MHz	30 KHz

Table 24 Commercial Area Bandwidth/SCS for Macro

Refer to the Appendix 10 - Commercial Area Macro Throughput Calculations

Calculating the Number of resource block (NRB):

$$\text{number of resource block} = \frac{(100 * 10^6 - 1.68 * 10^6)}{30 * 10^3 * 12} = 273$$

Calculating Peak throughput:

$$\boldsymbol{Cell\ peak\ throughput = \frac{(2.4192 * 10^6) * 273 * 4}{10^9} = 2.6417664\ Gbps}$$

Calculating User per cell:

$$\boldsymbol{User\ per\ cell = \frac{(2.6417664 * 10^3) * (100\% - 20\%)}{7.75} = 272.698}$$

Industrial Areas: Uses High demanded service like Live monitoring, Remote control equipment, IoT (Sensors, actuators), Automated vehicle. Live monitoring demands 5 Mbps, Remote control equipment uses 25 Mbps, IoT required 1 Mbps and Automated vehicle needs 10 Mbps.

Service	Bandwidth Required	Average demand (Mbps)
Live monitoring	5 Mbps	10.25
Remote control equipment	25 Mbps	
IoT (Sensors, actuators)	1 Mbps	
Automated vehicle	10 Mbps	

Table 25 Industrial Area Demamd

Industrial Areas uses macro cells.

Macro:

Since Industrial Areas demand massive amount of bandwidth for the listed services. The best choice for this high demand is 100 MHz for the bandwidth and the subcarrier spacing according to the 3GPP in Figure 8 FR1 will be 30 KHz to ensure low latency and speed for these services.

Bandwidth (MHz)	Subcarrier spacing (KHz)
100 MHz	30 KHz

Table 26 Industrial Area Bandwidth/SCS for Macro

Refer to the Appendix 11 – Industrial Area Macro Throughput Calculations

Calculating the Number of resource block (NRB):

$$\text{number of resource block} = \frac{(100 * 10^6 - 1.68 * 10^6)}{30 * 10^3 * 12} = 273$$

Calculating Peak throughput:

$$\boldsymbol{Cell\ peak\ throughput} = \frac{\left(\mathbf{2.4192} * \mathbf{10^6}\right) * \mathbf{273} * \mathbf{4}}{\mathbf{10^9}} = \mathbf{2.6417664}\ \boldsymbol{Gbps}$$

Calculating User per cell:

$$\boldsymbol{User\ per\ cell} = \frac{(\mathbf{2.6417664} * \mathbf{10^3}) * (\mathbf{100\%} - \mathbf{20\%})}{\mathbf{10.25}} = \mathbf{206.186}$$

Open Spaces: Open area includes services like web browsing, social media and video streaming. Social media and web browsing demands up to 5 Mbps and video streaming needs required approximately 10 Mbps.

Service	Bandwidth Required	Average demand (Mbps)
web browsing	5 Mbps	6.6
social media	5 Mbps	
video streaming	10 Mbps	

Table 27 Open Spaves Demand

Open Spaces will use macro cells only, there will be no micro cells.

Macro:

Bandwidth (MHz)	Subcarrier spacing (KHz)
100 MHz	30 KHz

Table 28 Open Spaces Bandwidth/SCS for Macro

According to 3GPP the 100 MHz bandwidth with a subcarrier spacing value 30 KHz is the standard for FR1 Spectrums.

Refer to the Appendix 12 – Open Spaces Macro Throughput Calculations

Calculating the Number of resource block (NRB):

$$\text{number of resource block} = \frac{(100 * 10^6 - 1.68 * 10^6)}{30 * 10^3 * 12} = 273$$

Calculating Peak throughput:

$$\boldsymbol{Cell\ peak\ throughput = \frac{(2.4192 * 10^6) * 273 * 4}{10^9} = 2.6417664\ Gbps}$$

Calculating User per cell:

$$\boldsymbol{User\ per\ cell = \frac{(2.6417664 * 10^3) * (100\% - 20\%)}{6.6} = 320.214}$$

Educational Area: The Common service used in educational areas is Web browsing, Cloud-based apps, Video conferencing and Smart classroom management systems. Web browsing demands 5 Mbps, Cloud-based apps use 10 Mbps, Video conferencing 10 Mbps and Smart classroom management systems (SSB) needs up to 5 Mbps.

Service	Bandwidth Required	Average demand (Mbps)
Web browsing	5 Mbps	15
Cloud-based apps	10 Mbps	
Video conferencing	10 Mbps	
Smart classroom management systems	5 Mbps	

Table 29 Educational Area Demand

Micro cells and macro cells are used in educational area.

Micro:

Bandwidth (MHz)	Subcarrier spacing (KHz)
200 MHz	60 KHz

Table 30 Educational Area Bandwidth/SCS for Micro

Since the educational area demand a lot of Mbps like the whole education area in Isa Town, the 200 MHz bandwidth and Subcarrier spacing value 60 KHz is much suitable for the FR2 spectrum range. Our selection for the 200 MHz and 60 SCS is to ensure balance between throughput and latency. In addition, this configuration will support as much user as possible at the same time.

Refer to the Appendix 13 – Educational Area Micro Throughput Calculations

Calculating the Number of resource block (NRB):

$$\text{number of resource block} = \frac{(200 * 10^6 - 1.68 * 10^6)}{60 * 10^3 * 12} = 275$$

Calculating Peak throughput:

$$\boldsymbol{Cell\ peak\ throughput} = \frac{(\mathbf{2.4192} * \mathbf{10^6}) * \mathbf{275} * \mathbf{4}}{\mathbf{10^9}} = \mathbf{2.66112}\ \boldsymbol{Gbps}$$

Calculating User per cell:

$$\boldsymbol{User\ per\ cell} = \frac{(\mathbf{2.66112} * \mathbf{10^3}) * (\mathbf{100\% - 20\%})}{\mathbf{15}} = \mathbf{141.926}$$

Macro:

Educational areas with macro cells such as UOB have wider area and less crowded compared to the other areas. Therefore, the suitable bandwidth and SCS will be 100 MHz and 30 KHz under FR1 spectrum frequency and the selected SCS will provide balance between latency and throughput.

Bandwidth (MHz)	Subcarrier spacing (KHz)
100 MHz	30 KHz

Table 31 Educatioanl Area Bandwidth/SCS for Macro

Refer to the Appendix 14 - Educational Area Macro Throughput Calculations

Calculating the Number of resource block (NRB):

$$\text{number of resource block} = \frac{(100 * 10^6 - 1.68 * 10^6)}{30 * 10^3 * 12} = 273$$

Calculating Peak throughput:

$$\boldsymbol{Cell\ peak\ throughput} = \frac{(\mathbf{2.4192} * \mathbf{10^6}) * \mathbf{273} * \mathbf{4}}{\mathbf{10^9}} = \mathbf{2.6417664}\ \boldsymbol{Gbps}$$

Calculating User per cell:

$$\boldsymbol{User\ per\ cell} = \frac{(\mathbf{2.6417664} * \mathbf{10^3}) * (\mathbf{100\% - 20\%})}{\mathbf{15}} = \mathbf{140.894}$$

Airport: The services used on airport areas are Digital boarding, Surveillance cameras, High speed public network, Web browsing & messaging. Surveillance cameras demand 15 Mbps, Digital boarding around 1 Mbps, High speed public network required 100 Mbps and Web browsing & messaging demands up to 5 Mbps.

Service	Bandwidth Required	Average demand (Mbps)
Web browsing	5 Mbps	15
Cloud-based apps	10 Mbps	
Video conferencing	10 Mbps	
Smart classroom management systems	5 Mbps	

Table 32 Airport Demand

Only micro cells will be used in airport

Micro:

Since the airport count as crowded area, it needs to have reliable network with lower latency and good throughput to serve a lot of people at the same time. To achieve that on an area like this we need to chose higher bandwidth like 200 with lower subcarrier spacing to ensure lower latency like 60 KHz.

Bandwidth (MHz)	Subcarrier spacing (KHz)
200 MHz	60 KHz

Table 33 Airport Bandwidth/SCS for Micro

Refer to the Appendix 15 – Airport Micro Throughput Calculation

Calculating the Number of resource block (NRB):

$$\text{number of resource block} = \frac{(200 * 10^6 - 1.68 * 10^6)}{60 * 10^3 * 12} = 275$$

Calculating Peak throughput:

$$\boldsymbol{Cell\ peak\ throughput} = \frac{\left(\mathbf{2.4192} * \mathbf{10^6}\right) * \mathbf{275} * \mathbf{4}}{\mathbf{10^9}} = \mathbf{2.66112}\ \boldsymbol{Gbps}$$

Calculating User per cell:

$$\boldsymbol{User\ per\ cell} = \frac{(\mathbf{2.66112} * \mathbf{10^3}) * (\mathbf{100\%} - \mathbf{20\%})}{\mathbf{15}} = \mathbf{141.926}$$

Network topology justification for various scenarios

Urban Areas (UMi Cells)

In **urban residential areas** such as Sanabis, Jurdab, and Sanad, where UMi (Urban Micro) cells are used, the population density is high. This makes sector antennas the ideal choice. Sector antennas reduce interference, provide more focused and controlled coverage, and allow frequency reuse, which helps in managing network load and improving performance. These antennas typically cover angles between 60° to 120° and are often installed on towers or rooftops to provide the best coverage in dense environments.

Figure 10 Sector Antenna

In Sanabis and Jurdab, 2 sector antennas are sufficient for each area. Sanad requires about 6 sector antennas. In the Muharraq Governorate, both Muharraq City and Bahrain International Airport need around 7 sector antennas each to handle the high traffic and user density. Arad and Al Hidd, due to their larger coverage needs, require about 19 sector antennas, while Qalali needs 5. In the Northern Governorate, Karranah and Budaiya each require 4 sector antennas. Hamad Town, which covers a large residential area, needs about 21 sector antennas. Isa Town in the Southern Governorate needs 9, Al Mazrowiah requires 6, and Riffa, which is a major urban area, needs around 48 antennas for complete coverage.

In urban **commercial areas** like Seef, sector antennas are also the best choice. These areas include shopping centers, business buildings, and parks where user density is high and data demand is constant. Sector antennas can provide high-capacity coverage and help reduce interference, especially in crowded places. For Seef, approximately 5 sector antennas are required. **Educational zones** like Bahrain Polytechnic also benefit from sector antennas due to the high number of users and devices on campus.

Suburban Areas (UMa Cells)

In suburban **residential areas**, where UMa cells are typically used, sector antennas remain a strong option. These areas usually have more open space and moderate population density. When sector antennas are mounted at heights of around 25 to 30 meters, they can provide wide-area coverage efficiently. In addition to sector antennas, directional antennas can be used at the user end, such as at homes or customer premises to improve signal reception, especially in fixed wireless access (FWA) setups. However, directional antennas are not used for base stations.

Figure 11 Directional Antenna

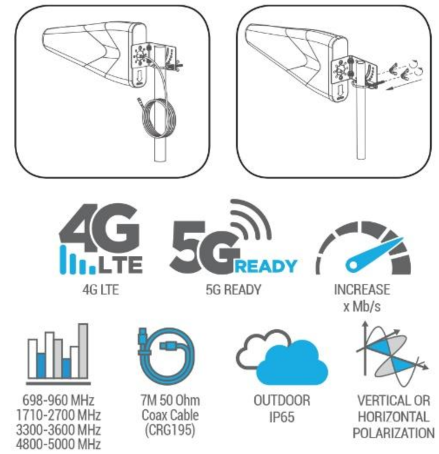


Figure 12 Directional Antenna Specifications

For example, Diyar Al Muharraq is a suburban **residential area** that requires about 9 sector antennas. In the Northern Governorate, Dar Kulaib only needs 1 sector antenna because of its smaller size, while Karzakan needs 2. In the Southern Governorate, Awali requires 2 sector antennas. The University of Bahrain, which is located in a suburban area, needs 2 sector antennas to cover its campus effectively.

In **industrial areas**, where user density is lower and towers are often spaced farther apart, omni-directional antennas are a better fit. These antennas provide 360-degree coverage around a single tower, making them cost-effective and easier to deploy in large open areas. They are particularly useful in zones where precise directional coverage isn't required and where users or connected devices are spread out. For instance, the Sitra Industrial Area needs around 10 omni-directional antennas, while Minaa Salman requires only 1. In the Muharraq Governorate, the Al Hidd Industrial Area needs 2 omni antennas and Khalifa Bin Salman Port requires about 10. In the Southern Industrial Area, a total of 15 omni-directional antennas are needed. Similarly, in **open spaces** and low-traffic areas like Um Al Nasan, 15 omni antennas are ideal because they offer broad coverage in all directions with simple setup and low maintenance.

Network implementation (screenshots of code)

Suburban Simulation Parameters

Following parameters will be used in the simulation for the macro cells specification:

Refer to Appendix 16, Appendix 17, Appendix 18, Appendix 19, Appendix 20, Appendix 21, Appendix 22, Appendix 23 to see the full code

- Total Area = 2720 m * 1360 m
- E-Node B Transmitting Power (Watt) = 50 Watt
- E-Node B Transmitting Power (dBm) = 46.9897 dBm
- Base Station Height = 35 m
- Carrier Frequency = 3.5 GHz
- Resource Blocks = 273
- Numerology index = 1
- Traffic simulated = VoIP, CBR
- User Equipment Transmitting Power = 26 Watt
- User Equipment Height = 2 m
- MIMO = 64
- Sampling Rate = 20 ms ≈ 0.02 ms
- Packet Per Second = 50 Packet Per Second

Refer to the appendix see the code for simulation

Simulation environment for macro cells:

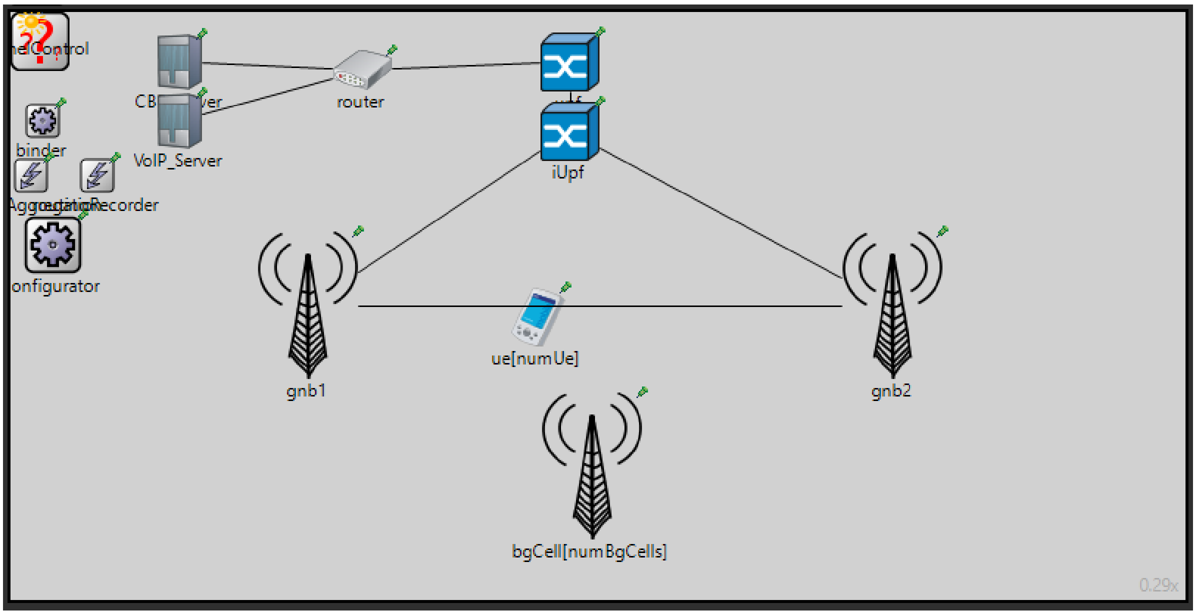


Figure 13 Suburban Simulation Environment

Urban Simulation Parameters:

Following parameters will be used in the simulation for the micro cells specification:

Refer to Appendix 24, Appendix 25, Appendix 26, Appendix 27, Appendix 28, Appendix 29, Appendix 30, Appendix 31 to see the full code

- Total Area = 1840 m * 920 m
- E-Node B Transmitting Power (Watt) = 100 Watt
- E-Node B Transmitting Power (dBm) = 50 dBm
- Base Station Height = 35 m
- Carrier Frequency = 24 GHz
- Resource Blocks = 275
- Numerology index = 2
- Traffic simulated = VoIP, CBR
- User Equipment Transmitting Power = 26 Watt
- User Equipment Height = 2 m
- MIMO = 32
- Sampling Rate = 20 ms ≈ 0.02 ms
- Packet Per Second = 50 Packet Per Second

Refer to the appendix see the code for simulation

Simulation environment for micro cells:

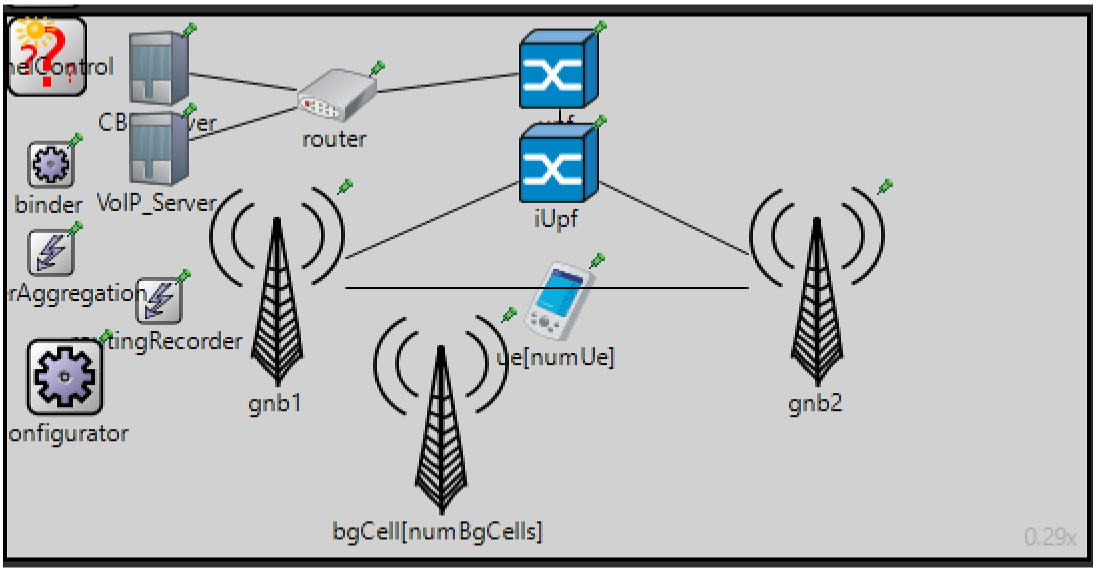


Figure 14 Urban Simulation Environment

Screenshots of results with discussion

CBR End-to-End Throughput:

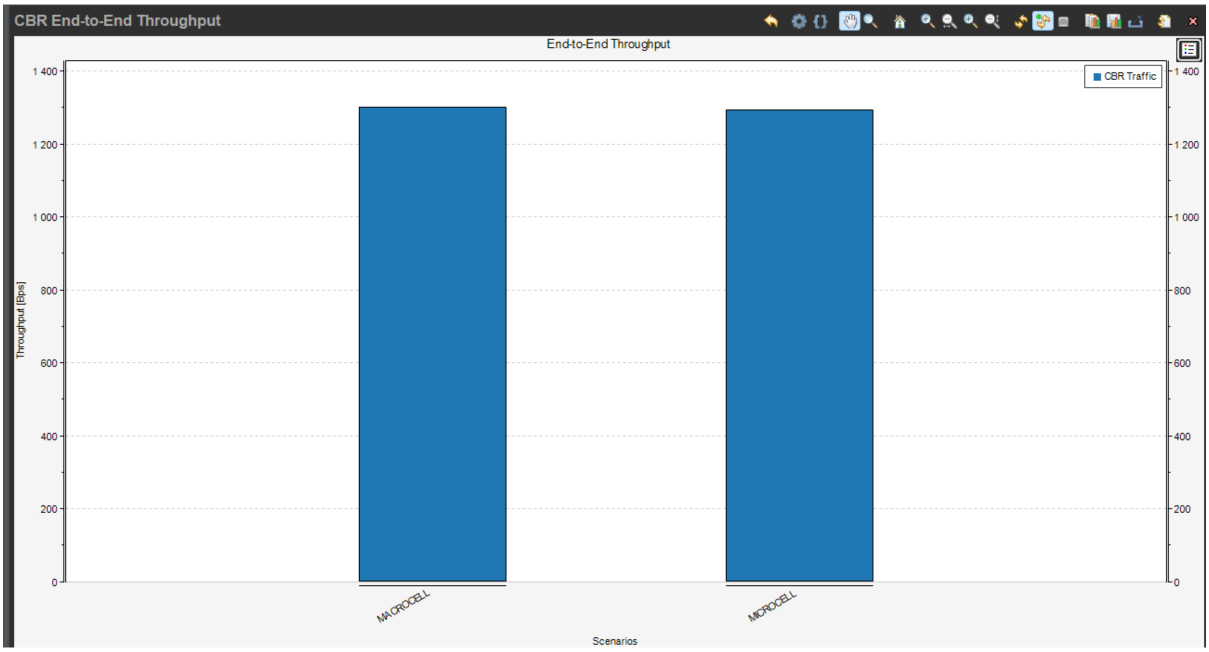


Figure 15 – CBR End-to-End Throughput

This bar chart explains the Constant Bit Rate CBR Throughput between Macro cell and Micro cell. The y-axis represents the Throughput in bps, and the x-axis represents the Macro and Micro cells. Both scenarios seem to have their average throughput close to 1300 bps, which suggests that both Macro and Micro cells performance for CBR throughputs are similar and either would qualify to be used.

CBR End-to-End Delay:

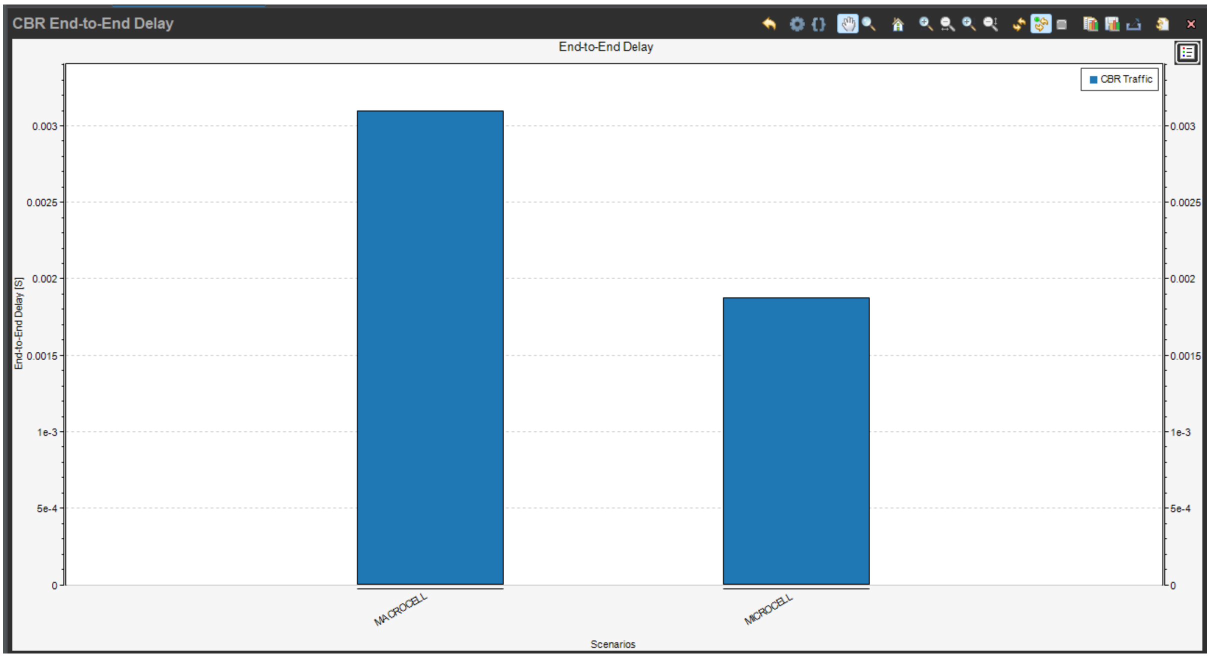


Figure 16 – CBR End-to-End Delay

The bar chart displays the Constant Bit Rate CBR end-to-end delay in two different scenarios of Macro cell and Microcell when it comes to handling CBR traffic. Macro cell seems to have a higher End-to-end Delay of around 0.003 seconds. Whereas, Microcell seems to have a lower CBR traffic delay of slightly less than 0.002 seconds. This may suggest that Microcell has lower CBR delays and performs better than Macro cell when it comes to CBR End-to-end Delay rate.

CBR End-to-End Packet loss:

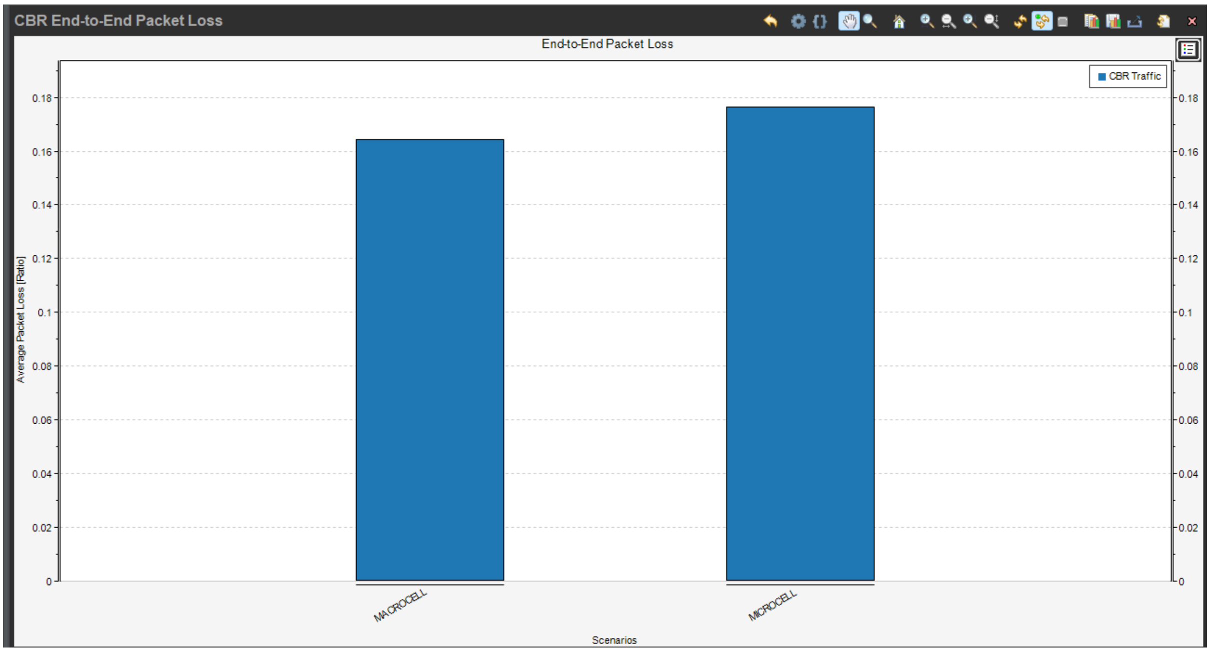


Figure 17 – CBR End-to-End Packet Loss

This bar chart displays the Constant Bit Rate CBR end-to-end packet loss for two different scenarios, the Macro and Micro cells. The y-axis demonstrates the Average packet loss ratio, and the x-axis demonstrates the two scenarios, the Macro and Micro cells. Macrocells seem to have slightly less than Microcells in terms of average packet loss Ratio, at around 0.16. whereas the microcells seem to have a slightly higher packet ratio loss of around 0.18. which may suggest that, in terms of CBR traffic and loss of packets, microcells have a slightly higher packet loss average ratio. In terms of performance, both will function properly with barely any noticeable change. To conclude with Microcells, experience less packet loss, indicating better reliability. This may be due to shorter transmission distances and fewer users competing for resources.

VoIP End-to-End Throughput:

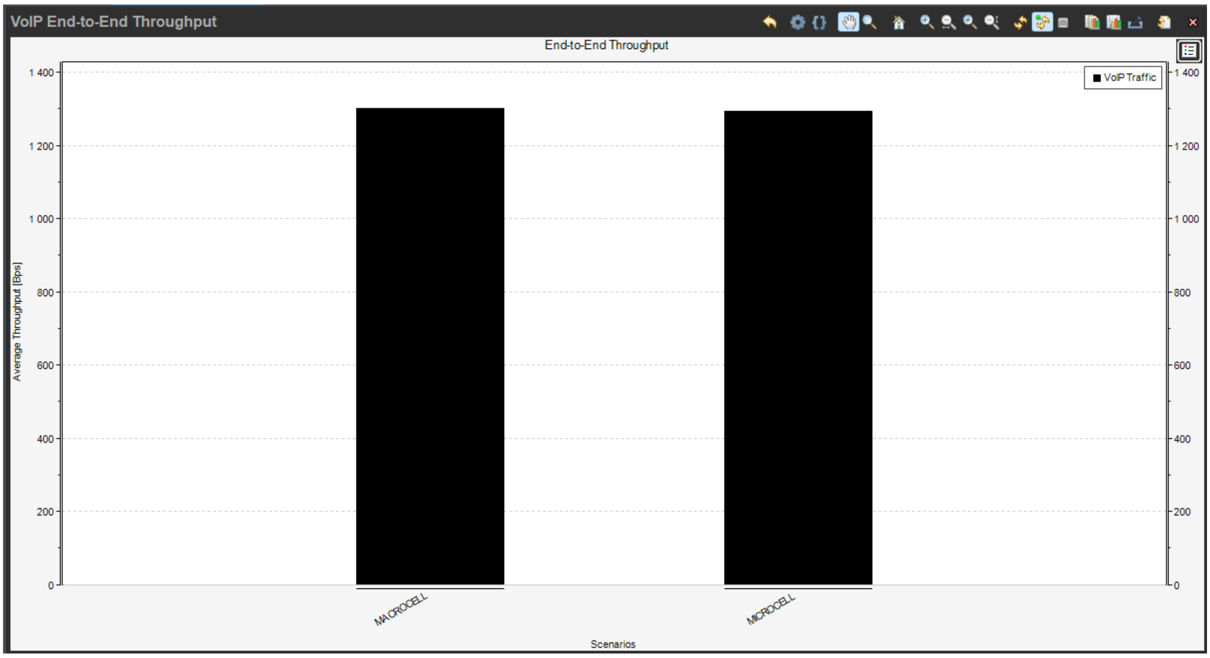


Figure 18 – VoIP End-to-End Throughput

The bar chart displays the Voice over IP VoIP End-to-End Throughput of two different scenarios, Macro cell and Microcells. Higher throughput in Microcells ensures better call quality and smoother communication. Both scenarios seem to have their average throughput close to 1300 Bps, which suggests that both MACROCELL and MICROCELL performance of VoIP throughputs are similar, and either would qualify to be used.

VoIP End-to-End Delay:

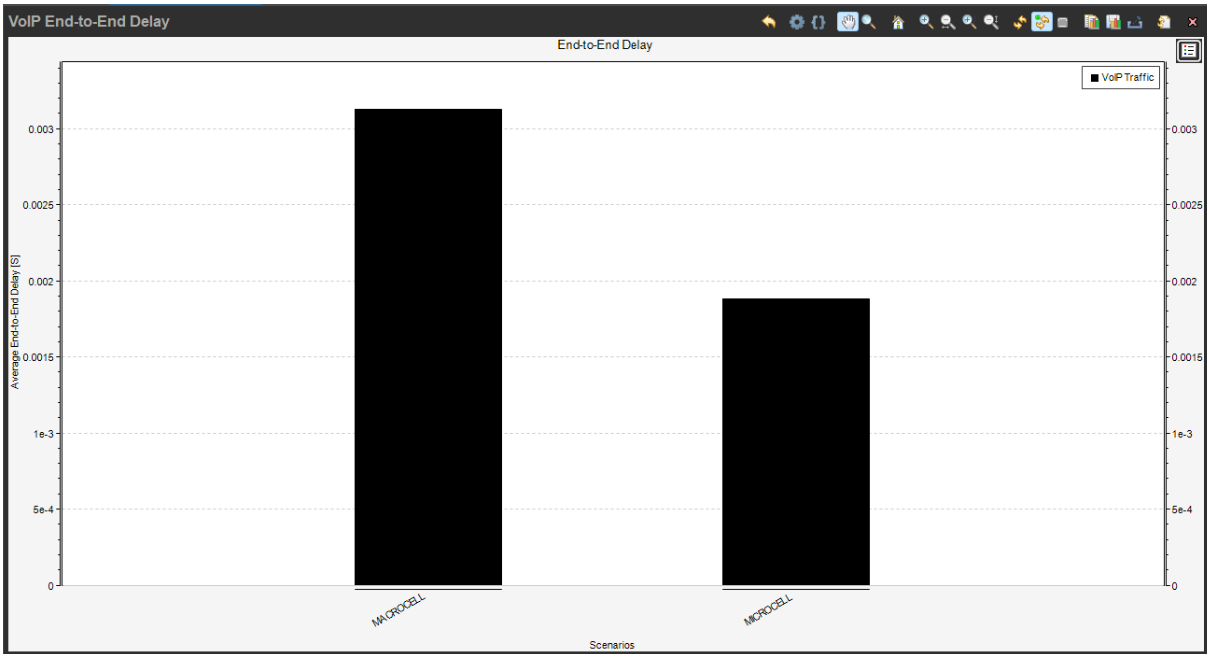


Figure 19 – VoIP End-to-End Delay

The Bar chart displays the End-to-end Delay of voice over IP (VoIP), where we have 2 scenarios of Macrocell and Microcell with different VoIP traffic. We notice that in the first scenario, the average

End-to-end Delay is the highest, with 0.003 seconds, which might indicate more delay. whereas the second scenario of Microcell shows a lower rate of delay, with approximately 0.002 seconds. This may suggest that Microcell transfers VoIP traffic better than Macro cell because the lower VoIP delay in Microcells ensures clearer, more immediate voice communication, reducing issues like echo and lag.

VoIP End-to-End Packet Loss:

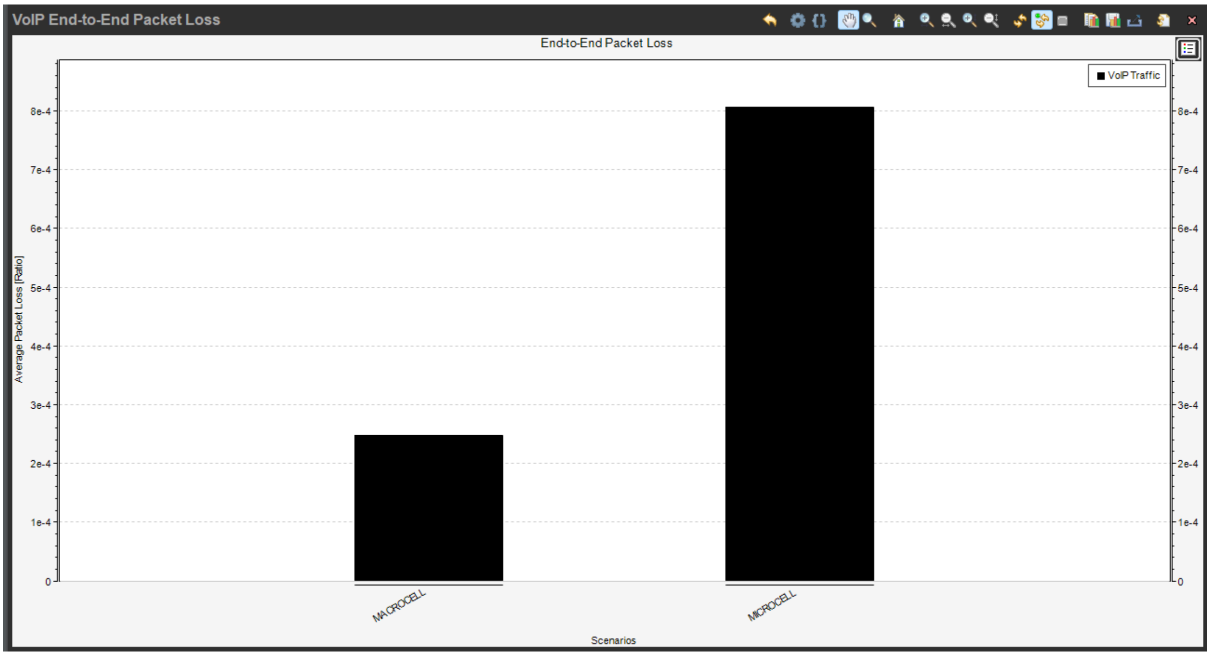
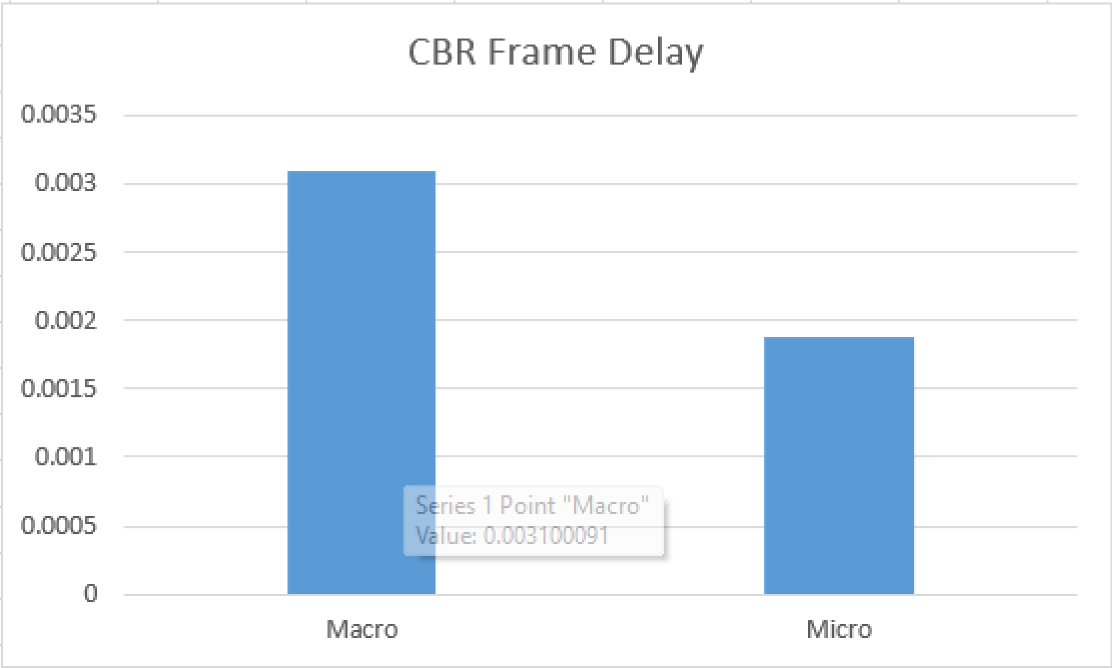


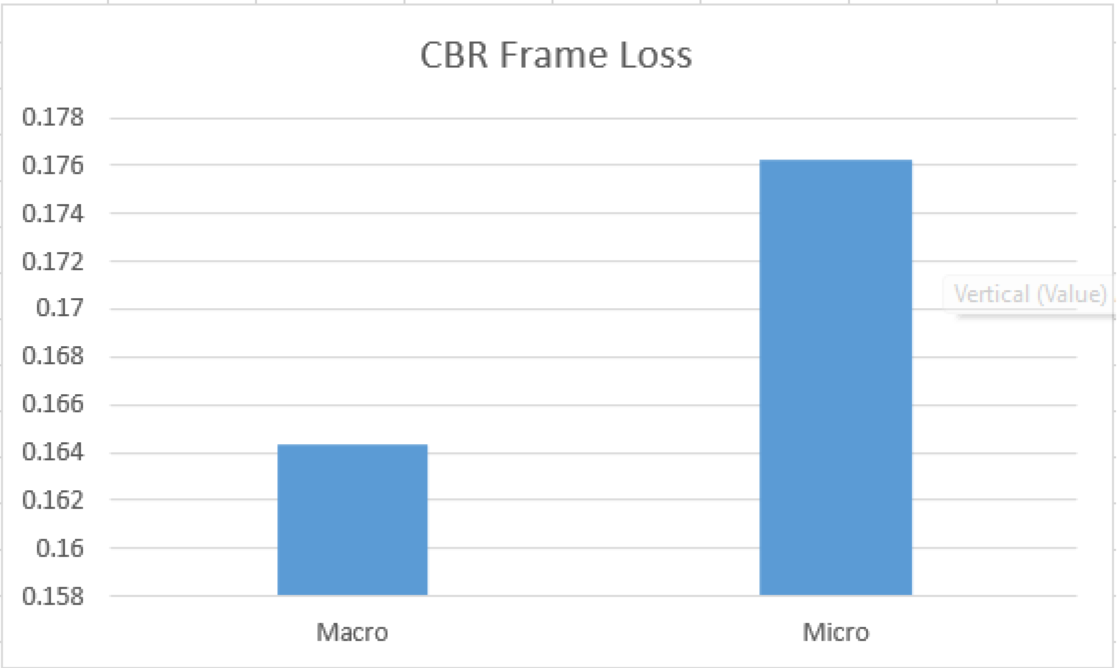
Figure 20 – VoIP End-to-End Packet Loss

The bar chart displays the voice over IP VoIP End-to-End Packet Loss for two different scenarios: Macro cell and Microcell. Macro cell seems to have slightly less packet loss than Microcell in terms of average packet loss ratio, at around 2.5 x 10-4. Whereas the Microcell scenario shows a slightly higher packet loss ratio of approximately 8 × 10-4. This may suggest that, in terms of VoIP traffic and packet loss, Microcell has a slightly higher average packet loss ratio. In terms of performance, Macro cell performs marginally better in maintaining lower packet loss rates. To sum Voice quality is very sensitive to packet loss. Microcells offer clearer calls and fewer dropped packets, making them more reliable for VoIP traffic.

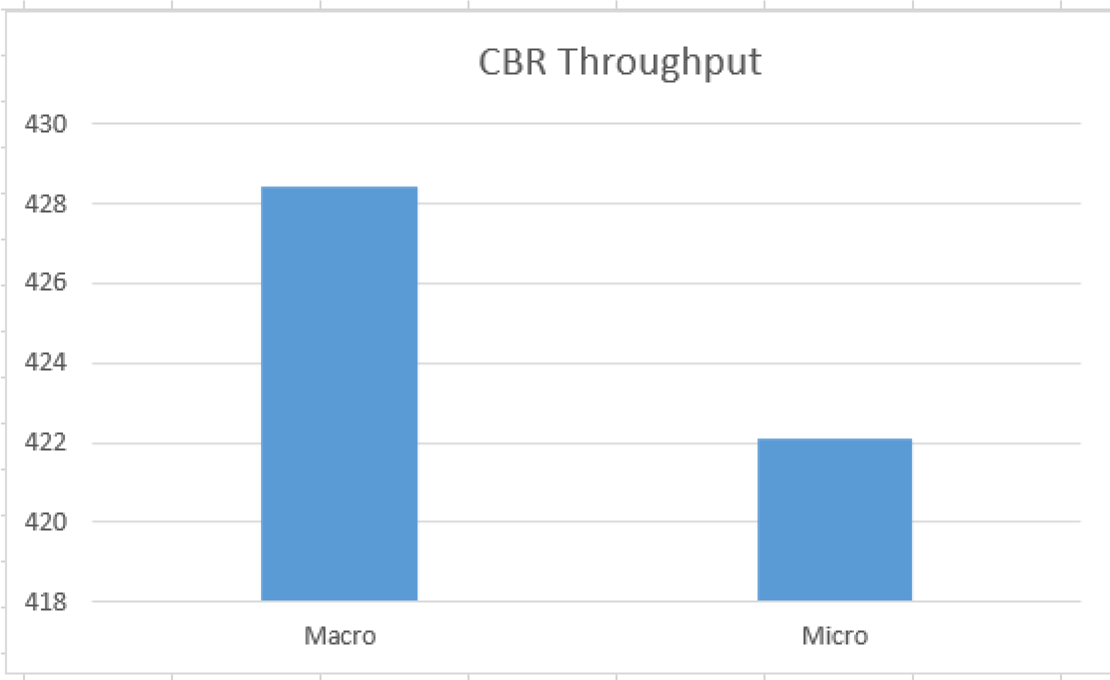
CBR Frame Delay Mean:



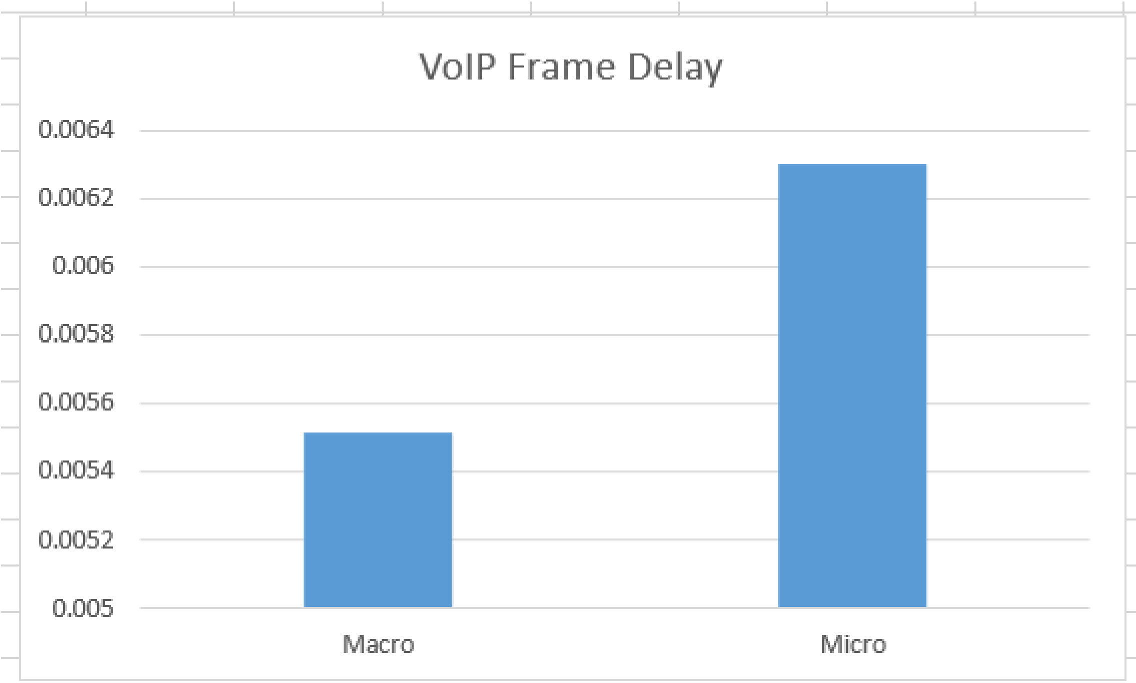
CBR Frame Loss Mean:



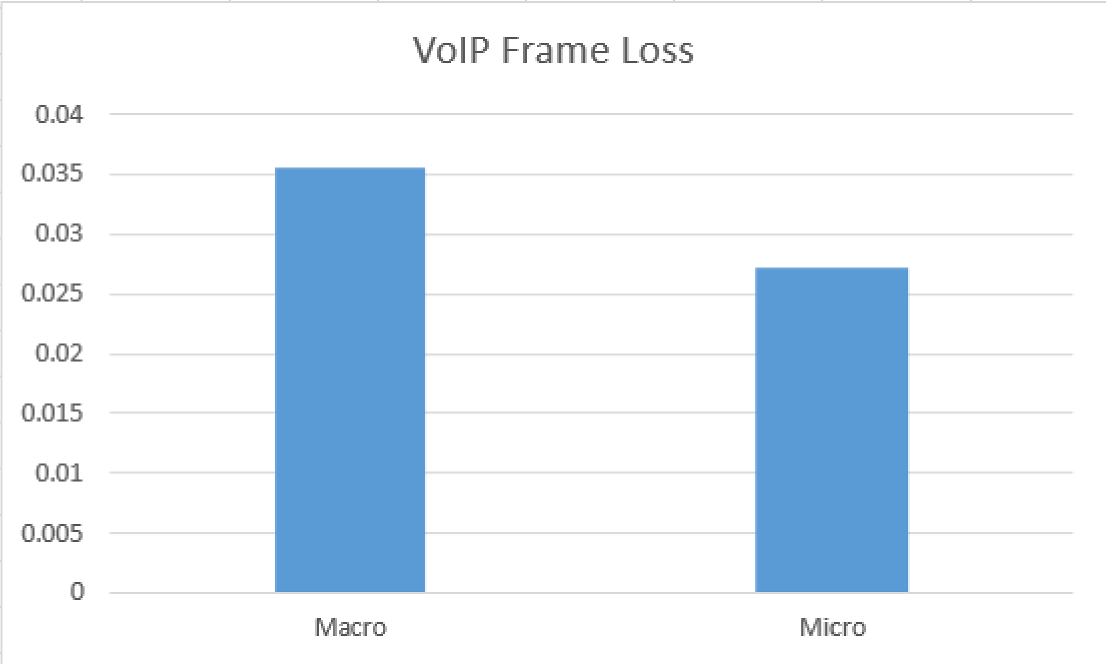
CBR Throughput Mean:



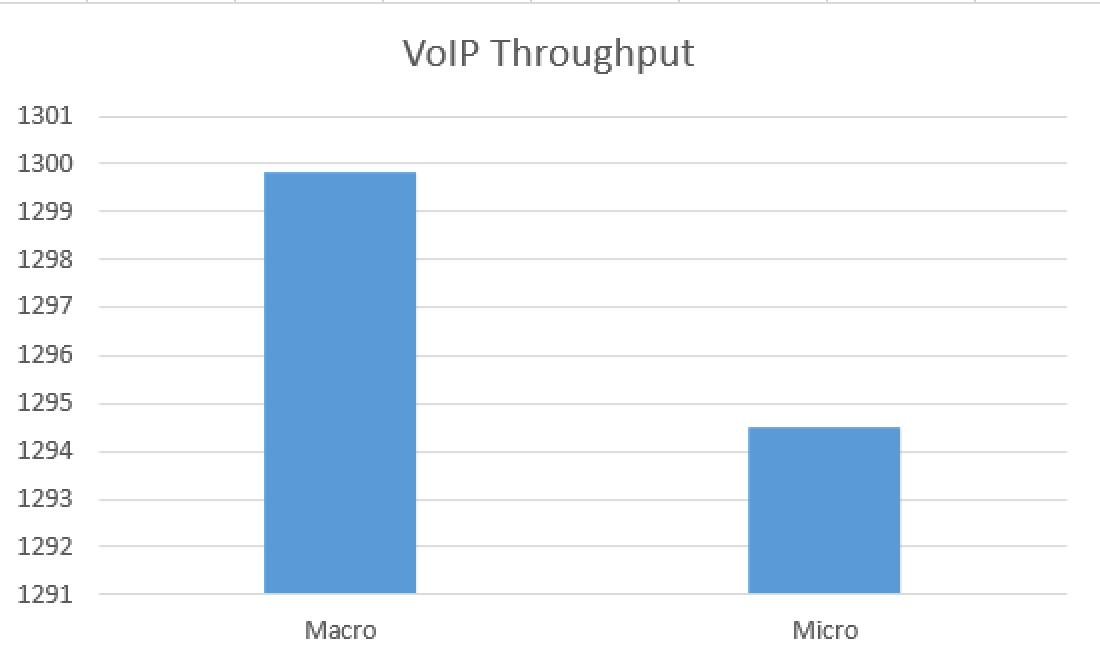
VoIP Frame Delay Mean:



VoIP Frame Loss Mean:



VoIP Throughput Mean:



Conclusion

This project has developed a plan for deploying 5G technology across Bahrain. With analysis and simulation, we demonstrated how 5G can improve network performance in urban and suburban areas. Our work provides a practical solution to enhance coverage, speed, and reliability for all users.

The results confirm that 5G is the right choice for Bahrain's growing connectivity needs. Hence, OKAKNet can build a network that supports businesses, government services, and everyday users. This project not only meets current demands but also prepares the infrastructure for tomorrow's technological advancements.

In summary, our 5G deployment plan offers a path to transform Bahrain's digital landscape, ensuring faster, more reliable internet for everyone.

Recommendations

1. **Prioritize High-Density Urban Areas First**
   High-density urban environments should be the top priority in 5G rollout plans due to the greater demand for data and connectivity.

2. **Use Macro Cells for Wide Suburban Coverage, Micro Cells for Congested Zones**
   In suburban regions with moderate population density, macro cells are suitable as they provide wider coverage with fewer sites. For congested zones within cities, such as business districts or stadiums, micro cells and small cells should be deployed to enhance capacity and reduce load on macro cells.

3. **Deploy Massive MIMO in Urban Areas**
   Massive MIMO technology significantly improves efficiency and user capacity. In urban areas, using antennas with configurations such as 8x8, 32x32 or higher helps serve more users simultaneously and reduces interference.

4. **Real-World Data and Simulations**
   Use actual population data, building density, terrain information and traffic patterns to simulate and plan antenna placements.

5. **Choose the Right Antenna Type Based on Area Characteristics**
   Antenna selection affects the signal distribution and network efficiency. For example, sector antennas in dense urban zones to control coverage and reduce interference. Omni-directional antennas in low-density or open areas like deserts or industrial zones. Use directional antennas at user premises areas to improve reception.

6. **Maintain Optimal Received Signal Power**
   Ensure that the signal strength at the user end remains within ideal thresholds like -70 dBm. This balances strong coverage and minimizes interference.

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Appendix

UMa Cells Calculations

Pt	50	Watt	in dBm	46.9897
Frequency	3.5	GHz		
BS Height	35	Meter		
UE Height	2	Meter		
MIMO	64	number of antennas		

Appendix 1 - UMa Parameters

Distance	Path Loss	Power Received
10	63.68197888	1.369520855
20	75.44623111	-10.39473138
30	82.32787752	-17.27637778
40	87.21048335	-22.15898361
50	90.99772665	-25.94622691
60	94.09212975	-29.04063001
70	96.70841029	-31.65691055
80	98.97473558	-33.92323584
90	100.9737762	-35.92227641
100	102.7619789	-37.71047914
110	104.379605	-39.32810528
120	105.856382	-40.80488224
130	107.2148851	-42.16338535
140	108.4726625	-43.42116278
150	109.6436253	-44.59212555
160	110.7389878	-45.68748807
170	111.7679227	-46.71642299
180	112.7380284	-47.68652864
190	113.6556696	-48.60416987
200	114.5262311	-49.47473138
210	115.3543089	-50.30280918
220	116.1438573	-51.09235751
230	116.8983027	-51.84680298
240	117.6206342	-52.56913447
250	118.3134744	-53.26197468
260	118.9791373	-53.92763758
270	119.6196748	-54.56817505
280	120.2369147	-55.18541501
290	120.8324926	-55.7809929
300	121.4078775	-56.35637778
310	121.9643939	-56.91289414
320	122.50324	-57.4517403
330	123.0255037	-57.97400391

340	123.532175	-58.48067522
350	124.0241581	-58.97265832
360	124.5022806	-59.45078087
370	124.9673023	-59.91580252
380	125.4199218	-60.3684221
390	125.8607837	-60.80928399
400	126.2904833	-61.23898361
410	126.709572	-61.65807227
420	127.1185612	-62.06706141
430	127.5179261	-62.46642639
440	127.9081095	-62.85660974
450	128.2895239	-63.23802418
460	128.6625549	-63.61105521
470	129.0275632	-63.97606343
480	129.3848864	-64.3333867
490	129.7348417	-64.68334195
500	130.0777267	-65.02622691
510	130.4138214	-65.36232163
520	130.7433896	-65.69188981
530	131.0666799	-66.01518013
540	131.383927	-66.33242728
550	131.6953528	-66.64385305
560	132.001167	-66.94966724
570	132.3015682	-67.2500685
580	132.5967449	-67.54524513
590	132.8868755	-67.83537576
600	133.1721297	-68.12063001
610	133.4526688	-68.4011691
620	133.7286461	-68.67714637
630	134.0002076	-68.94870782
640	134.2674923	-69.21599253
650	134.5306329	-69.47913312
660	134.7897559	-69.73825615
670	135.0449822	-69.99348243
680	135.2964272	-70.24492745
690	135.5442014	-70.49270161
700	135.7884103	-70.73691055
710	136.0291552	-70.97765541
720	136.2665328	-71.21503311
730	136.5006363	-71.44913652
740	136.7315545	-71.68005475
750	136.9593731	-71.90787332
760	137.1841741	-72.13267433
770	137.4060364	-72.35453668

780	137.625036	-72.57353622
790	137.8412456	-72.78974587
800	138.0547356	-73.00323584
810	138.2655734	-73.21407368
820	138.4738242	-73.4223245
830	138.6795507	-73.62805099
840	138.8828134	-73.83131364
850	139.0836705	-74.03217076
860	139.2821784	-74.23067862
870	139.4783913	-74.42689154
880	139.6723617	-74.62086197
890	139.8641403	-74.8126406
900	140.0537762	-75.00227641
910	140.2413165	-75.18981676
920	140.4268072	-75.37530744
930	140.6102925	-75.55879277
940	140.7918154	-75.74031566
950	140.9714174	-75.91991764
960	141.1491387	-76.09763893
970	141.3250183	-76.27351852
980	141.4990939	-76.44759418
990	141.6714023	-76.61990255
1000	141.8419789	-76.79047914

Appendix 2 – UMa Path Loss and Power received

One cell coverage	1.452672443

Appendix 3 – UMa One Cell Coverage

UMi Cells Calculation

Pt	100	Watt	in dBm	50
Frequency	24	GHz		
BS Height	35	Meter		
UE Height	2	Meter		
MIMO	128	number of antennas		

Appendix 4 – UMi Parameters

Distance	Path Loss	Power Received
10	80.40484283	-9.332743135
20	92.16909506	-21.09699537
30	99.05074147	-27.97864177
40	103.9333473	-32.8612476
50	107.7205906	-36.6484909
60	110.8149937	-39.742894
70	113.4312742	-42.35917454
80	115.6975995	-44.62549983
90	117.6966401	-46.6245404

100	119.4848428	-48.41274314
110	121.102469	-50.03036927
120	122.5792459	-51.50714623
130	123.937749	-52.86564934
140	125.1955265	-54.12342677
150	126.3664892	-55.29438954
160	127.4618518	-56.38975206
170	128.4907867	-57.41868698
180	129.4608923	-58.38879263
190	130.3785336	-59.30643386
200	131.2490951	-60.17699537
210	132.0771729	-61.00507317
220	132.8667212	-61.7946215
230	133.6211667	-62.54906697
240	134.3434982	-63.27139846
250	135.0363384	-63.96423867
260	135.7020013	-64.62990157
270	136.3425387	-65.27043904
280	136.9597787	-65.887679
290	137.5553566	-66.48325689
300	138.1307415	-67.05864177
310	138.6872578	-67.61515813
320	139.226104	-68.15400429
330	139.7483676	-68.67626791
340	140.2550389	-69.18293921
350	140.747022	-69.67492231
360	141.2251446	-70.15304487
370	141.6901662	-70.61806651
380	142.1427858	-71.07068609
390	142.5836477	-71.51154798
400	143.0133473	-71.9412476
410	143.432436	-72.36033626
420	143.8414251	-72.7693254
430	144.2407901	-73.16869038
440	144.6309734	-73.55887373
450	145.0123879	-73.94028817
460	145.3854189	-74.3133192
470	145.7504271	-74.67832742
480	146.1077504	-75.03565069
490	146.4577056	-75.38560594
500	146.8005906	-75.7284909
510	147.1366853	-76.06458562
520	147.4662535	-76.3941538
530	147.7895438	-76.71744412

540	148.106791	-77.03469127
550	148.4182167	-77.34611704
560	148.7240309	-77.65193123
570	149.0244322	-77.95233249
580	149.3196088	-78.24750912
590	149.6097394	-78.53763975
600	149.8949937	-78.822894
610	150.1755328	-79.10343309
620	150.4515101	-79.37941036
630	150.7230715	-79.65097181
640	150.9903562	-79.91825652
650	151.2534968	-80.18139711
660	151.5126198	-80.44052014
670	151.7678461	-80.69574642
680	152.0192911	-80.94719144
690	152.2670653	-81.1949656
700	152.5112742	-81.43917454
710	152.7520191	-81.6799194
720	152.9893968	-81.9172971
730	153.2235002	-82.15140051
740	153.4544184	-82.38231874
750	153.682237	-82.61013731
760	153.907038	-82.83493832
770	154.1289004	-83.05680067
780	154.3478999	-83.27580021
790	154.5641096	-83.49200986
800	154.7775995	-83.70549983
810	154.9884374	-83.91633767
820	155.1966882	-84.12458849
830	155.4024147	-84.33031499
840	155.6056773	-84.53357763
850	155.8065344	-84.73443475
860	156.0050423	-84.93294261
870	156.2012552	-85.12915553
880	156.3952257	-85.32312596
890	156.5870043	-85.51490459
900	156.7766401	-85.7045404
910	156.9641804	-85.89208075
920	157.1496711	-86.07757143
930	157.3331565	-86.26105676
940	157.5146794	-86.44257965
950	157.6942813	-86.62218163
960	157.8720026	-86.79990292
970	158.0478822	-86.97578251

980	158.2219579	-87.14985817
990	158.3942662	-87.32216654
1000	158.5648428	-87.49274314

Appendix 5 – UMi Path Loss and Power received

One cell coverage	0.664761005

Appendix 6 – UMi One Cell Coverage

Throughput Calculation

Peak System Throughput for residental area		
Important Parameters		
Channel Bandwidth (BW)	200	MHz
Subcarrier Spacing (SCS)	60	KHz
SE=Modulation Order×Coding Rate	7.2	bps/Hz, QAM-256 it's highest modulation scheme practically used and it is a most important parameter.
Number of Resource Blocks (NRB)	275	NRB= BW/SCS
Number of Layers (NL)	4	
Symbols per Slot	14	(typically 14 for normal CP
Overhead (OH)		
Calculate Resource Block Capacity		
Subcarriers per Resource Block	12	
Symbols per Slot	14	
Capacity per RB per Slot	1209.6	bits/slot
Slot duration	8.33E-06	microseconds
Calculate Throughput per Resource Block		
Throughput per RB=Capacity per RB per Slot×Number of Slots per Second		
Number of Slots per Second	2000	
Throughput per RB	2.4192	**Mbps**
Calculate Total Cell Peak Throughput		
Total Cell Peak Throughput=Throughput per RB×Number of Resource Blocks×Number of Layers		
Total Cell Peak Throughput	2.66112	**Gbps**
Average user demand	13	
Total users per cell	163.7612308	

Appendix 7 - Residential Area Micro Throughput Calculations

Peak System Throughput for residental area		
Important Parameters		
Channel Bandwidth (BW)	100	MHz
Subcarrier Spacing (SCS)	30	KHz
SE=Modulation Order×Coding Rate	7.2	bps/Hz, QAM-256 it's highest modulation scheme practically used and it is a most important parameter.
Number of Resource Blocks (NRB)	273	NRB= BW/SCS
Number of Layers (NL)	4	
Symbols per Slot	14	(typically 14 for normal CP
Overhead (OH)		
Calculate Resource Block Capacity		
Subcarriers per Resource Block	12	
Symbols per Slot	14	
Capacity per RB per Slot	1209.6	bits/slot
Slot duration	1.67E-05	microseconds
Calculate Throughput per Resource Block		
Throughput per RB=Capacity per RB per Slot×Number of Slots per Second		
Number of Slots per Second	2000	
Throughput per RB	2.4192	**Mbps**
Calculate Total Cell Peak Throughput		
Total Cell Peak Throughput=Throughput per RB×Number of Resource Blocks×Number of Layers		
Total Cell Peak Throughput	2.6417664	**Gbps**
Average user demand	13	
Total users per cell	162.57024	

Appendix 8 - Residential Area Macro Throughput Calculations

Peak System Throughput for Commercial areas		
Important Parameters		
Channel Bandwidth (BW)	200	MHz
Subcarrier Spacing (SCS)	60	KHz
SE=Modulation Order×Coding Rate	7.2	bps/Hz, QAM-256 it's highest modulation scheme practically used and it is a most important parameter.
Number of Resource Blocks (NRB)	275	NRB= BW/SCS
Number of Layers (NL)	4	
Symbols per Slot	14	(typically 14 for normal CP
Overhead (OH)		
Calculate Resource Block Capacity		
Subcarriers per Resource Block	12	
Symbols per Slot	14	
Capacity per RB per Slot	1209.6	bits/slot
Slot duration	8.33E-06	microseconds
Calculate Throughput per Resource Block		
Throughput per RB=Capacity per RB per Slot×Number of Slots per Second		
Number of Slots per Second	2000	
Throughput per RB	2.4192	**Mbps**
Calculate Total Cell Peak Throughput		
Total Cell Peak Throughput=Throughput per RB×Number of Resource Blocks×Number of Layers		
Total Cell Peak Throughput	2.66112	**Gbps**
Average user demand	7.75	
Total users per cell	274.6962581	

Appendix 9 - Commercial Area Micro Throughput Calculations

Peak System Throughput for Commercial areas		
Important Parameters		
Channel Bandwidth (BW)	100	MHz
Subcarrier Spacing (SCS)	30	KHz
SE=Modulation Order×Coding Rate	7.2	bps/Hz, QAM-256 it's highest modulation scheme practically used and it is a most important parameter.
Number of Resource Blocks (NRB)	273	NRB= BW/SCS
Number of Layers (NL)	4	
Symbols per Slot	14	(typically 14 for normal CP
Overhead (OH)		
Calculate Resource Block Capacity		
Subcarriers per Resource Block	12	
Symbols per Slot	14	
Capacity per RB per Slot	1209.6	bits/slot
Slot duration	1.67E-05	microseconds
Calculate Throughput per Resource Block		
Throughput per RB=Capacity per RB per Slot×Number of Slots per Second		
Number of Slots per Second	2000	
Throughput per RB	2.4192	**Mbps**
Calculate Total Cell Peak Throughput		
Total Cell Peak Throughput=Throughput per RB×Number of Resource Blocks×Number of Layers		
Total Cell Peak Throughput	2.6417664	**Gbps**
Average user demand	7.75	
Total users per cell	272.6984671	

Appendix 10 - Commercial Area Macro Throughput Calculations

Peak System Throughput for Industrial areas		
Important Parameters		
Channel Bandwidth (BW)	100	MHz
Subcarrier Spacing (SCS)	30	KHz
SE=Modulation Order×Coding Rate	7.2	bps/Hz, QAM-256 it's highest modulation scheme practically used and it is a most important parameter.
Number of Resource Blocks (NRB)	273	NRB= BW/SCS
Number of Layers (NL)	4	
Symbols per Slot	14	(typically 14 for normal CP
Overhead (OH)		
Calculate Resource Block Capacity		
Subcarriers per Resource Block	12	
Symbols per Slot	14	
Capacity per RB per Slot	1209.6	bits/slot
Slot duration	1.67E-05	microseconds
Calculate Throughput per Resource Block		
Throughput per RB=Capacity per RB per Slot×Number of Slots per Second		
Number of Slots per Second	2000	
Throughput per RB	2.4192	**Mbps**
Calculate Total Cell Peak Throughput		
Total Cell Peak Throughput=Throughput per RB×Number of Resource Blocks×Number of Layers		
Total Cell Peak Throughput	2.6417664	**Gbps**
Average user demand	15	
Total users per cell	140.894208	

Appendix 11 – Industrial Area Macro Throughput Calculations

Peak System Throughput for Open Spaces		
Important Parameters		
Channel Bandwidth (BW)	100	MHz
Subcarrier Spacing (SCS)	30	KHz
SE=Modulation Order×Coding Rate	7.2	bps/Hz, QAM-256 it's highest modulation scheme practically used and it is a most important parameter.
Number of Resource Blocks (NRB)	273	NRB= BW/SCS
Number of Layers (NL)	4	
Symbols per Slot	14	(typically 14 for normal CP
Overhead (OH)		
Calculate Resource Block Capacity		
Subcarriers per Resource Block	12	
Symbols per Slot	14	
Capacity per RB per Slot	1209.6	bits/slot
Slot duration	1.67E-05	microseconds
Calculate Throughput per Resource Block		
Throughput per RB=Capacity per RB per Slot×Number of Slots per Second		
Number of Slots per Second	2000	
Throughput per RB	2.4192	**Mbps**
Calculate Total Cell Peak Throughput		
Total Cell Peak Throughput=Throughput per RB×Number of Resource Blocks×Number of Layers		
Total Cell Peak Throughput	2.6417664	**Gbps**
Average user demand	6.6	
Total users per cell	320.2141091	

Appendix 12 – Open Spaces Macro Throughput Calculations

Peak System Throughput for Educational Areas		
Important Parameters		
Channel Bandwidth (BW)	200	MHz
Subcarrier Spacing (SCS)	60	KHz
SE=Modulation Order×Coding Rate	7.2	bps/Hz, QAM-256 it's highest modulation scheme practically used and it is a most important parameter.
Number of Resource Blocks (NRB)	275	NRB= BW/SCS
Number of Layers (NL)	4	
Symbols per Slot	14	(typically 14 for normal CP
Overhead (OH)		
Calculate Resource Block Capacity		
Subcarriers per Resource Block	12	
Symbols per Slot	14	
Capacity per RB per Slot	1209.6	bits/slot
Slot duration	8.33E-06	microseconds
Calculate Throughput per Resource Block		
Throughput per RB=Capacity per RB per Slot×Number of Slots per Second		
Number of Slots per Second	2000	
Throughput per RB	2.4192	**Mbps**
Calculate Total Cell Peak Throughput		
Total Cell Peak Throughput=Throughput per RB×Number of Resource Blocks×Number of Layers		
Total Cell Peak Throughput	2.66112	**Gbps**
Average user demand	15	
Total users per cell	141.9264	Educational Areas

Appendix 13 – Educational Area Micro Throughput Calculations

Peak System Throughput for Educational Areas		
Important Parameters		
Channel Bandwidth (BW)	100	MHz
Subcarrier Spacing (SCS)	30	KHz
SE=Modulation Order×Coding Rate	7.2	bps/Hz, QAM-256 it's highest modulation scheme practically used and it is a most important parameter.
Number of Resource Blocks (NRB)	273	NRB= BW/SCS
Number of Layers (NL)	4	
Symbols per Slot	14	(typically 14 for normal CP
Overhead (OH)		
Calculate Resource Block Capacity		
Subcarriers per Resource Block	12	
Symbols per Slot	14	
Capacity per RB per Slot	1209.6	bits/slot
Slot duration	1.67E-05	microseconds
Calculate Throughput per Resource Block		
Throughput per RB=Capacity per RB per Slot×Number of Slots per Second		
Number of Slots per Second	2000	
Throughput per RB	2.4192	**Mbps**
Calculate Total Cell Peak Throughput		
Total Cell Peak Throughput=Throughput per RB×Number of Resource Blocks×Number of Layers		
Total Cell Peak Throughput	2.6417664	**Gbps**
Average user demand	15	
Total users per cell	140.894208	

Appendix 14 - Educational Area Macro Throughput Calculations

Peak System Throughput for Airport		
Important Parameters		
Channel Bandwidth (BW)	200	MHz
Subcarrier Spacing (SCS)	60	KHz
SE=Modulation Order×Coding Rate	7.2	bps/Hz, QAM-256 it's highest modulation scheme practically used and it is a most important parameter.
Number of Resource Blocks (NRB)	275	NRB= BW/SCS
Number of Layers (NL)	4	
Symbols per Slot	14	(typically 14 for normal CP
Overhead (OH)		
Calculate Resource Block Capacity		
Subcarriers per Resource Block	12	
Symbols per Slot	14	
Capacity per RB per Slot	1209.6	bits/slot
Slot duration	8.33E-06	microseconds
Calculate Throughput per Resource Block		
Throughput per RB=Capacity per RB per Slot×Number of Slots per Second		
Number of Slots per Second	2000	
Throughput per RB	2.4192	**Mbps**
Calculate Total Cell Peak Throughput		
Total Cell Peak Throughput=Throughput per RB×Number of Resource Blocks×Number of Layers		
Total Cell Peak Throughput	2.66112	**Gbps**
Average user demand	15	
Total users per cell	141.9264	

Appendix 15 – Airport Micro Throughput Calculation

Omnet++ Code - Micro:

```
################ General PHY parameters ###############
**.downlink_interference = true
**.uplink_interference = true
**.ueTxPower = 26dBm
**.eNodeBTxPower = 50dBm
**.targetBler = 0.01
**.blerShift = 5
**.fbPeriod = 40tti   # reports CQI every 40ms
```

Appendix 16 – Micro Cells General Parameters

```
############## Floorplan parameters #################
**.mobility.constraintAreaMaxX = 1840m
**.mobility.constraintAreaMaxY = 920m
**.mobility.constraintAreaMinX = 0m
**.mobility.constraintAreaMinY = 0m
**.mobility.constraintAreaMinZ = 0m
**.mobility.constraintAreaMaxZ = 0m
**.mobility.initFromDisplayString = false
```

Appendix 17 – Micro Cells Floorplan Parameters

```
############### CA configuration #################
*.carrierAggregation.numComponentCarriers = 1
*.carrierAggregation.componentCarrier[0].numerologyIndex = ${u=2}
*.carrierAggregation.componentCarrier[0].carrierFrequency = 24GHz
*.carrierAggregation.componentCarrier[0].numBands = ${numRbs=275}

*.gnb*.cellularNic.numCarriers = 1
*.gnb*.cellularNic.channelModel[0].componentCarrierIndex = 0
*.ue[*].cellularNic.numCarriers = 1
*.ue[*].cellularNic.nrChannelModel[0].componentCarrierIndex = 0
```

Appendix 18 – Micro Cells CA Configurations

```
################ BS position #################
*.gnb1.mobility.initialX = 460m
*.gnb1.mobility.initialY = 460m
*.gnb2.mobility.initialX = 1380m
*.gnb2.mobility.initialY = 460m
```

Appendix 19 – Micro Cells BS Parameters

```
############### UE configuration ###################
*.numUe = ${numUEs=142}
```

Appendix 20 – Micro Cells Number of UE

```
############# UEs position ##################
*.ue[*].mobility.typename = "LinearMobility"
*.ue[*].mobility.initialX = uniform(0m,1840m)
*.ue[*].mobility.initialY = uniform(0m,920m)
*.ue[*].mobility.speed = uniform(3mps,13.89mps)
#-----------------------------------#
```

Appendix 21 – Micro Cells UE Position

```
#============ Application Setup ============
*.ue[*].app[0].typename = "CbrReceiver"
*.ue[*].app[0].localPort = 3000

*.CBR_Server.app[*].typename = "CbrSender"
*.CBR_Server.app[*].PacketSize = 1000
*.CBR_Server.app[*].sampling_time = 0.02s
*.CBR_Server.app[*].destAddress = "ue[" + string(ancestorIndex(0)) + "]"
*.CBR_Server.app[*].destPort = 3000
*.CBR_Server.app[*].localPort = 3088+ancestorIndex(0)
*.CBR_Server.app[*].startTime = uniform(0s,0.02s)
*.CBR_Server.app[*].finishTime = uniform(1199.9s,1199.92s)
```

Appendix 22 – Micro Cells CBR Parameters

```
##============= Application Setup =============
*.ue[*].app[1].typename = "VoIPReceiver"
*.ue[*].app[1].localPort = 4000

*.VoIP_Server.app[*].PacketSize = 100
*.VoIP_Server.app[*].destAddress = "ue[" + string(ancestorIndex(0)) + "]"
*.VoIP_Server.app[*].destPort = 4000
*.VoIP_Server.app[*].localPort = 4088+ancestorIndex(0)
*.VoIP_Server.app[*].typename = "VoIPSender"
*.VoIP_Server.app[*].startTime = uniform(0s,0.02s)
*.VoIP_Server.app[*].finishTime = uniform(1199.9s,1199.92s)

#-----------------------------------#
```

Appendix 23 – Micro Cells VoIP Parameters

Omnet++ Code - Macro:

```
################ General PHY parameters ###############
**.downlink_interference = true
**.uplink_interference = true
**.ueTxPower = 26dBm
**.eNodeBTxPower = 46.9897dBm
**.targetBler = 0.01
**.blerShift = 5
**.fbPeriod = 40tti    # reports CQI every 40ms
```

Appendix 24 - Macro Cells General Parameters

```
############### Floorplan parameters #################
**.mobility.constraintAreaMaxX = 2720m
**.mobility.constraintAreaMaxY = 1360m
**.mobility.constraintAreaMinX = 0m
**.mobility.constraintAreaMinY = 0m
**.mobility.constraintAreaMinZ = 0m
**.mobility.constraintAreaMaxZ = 0m
**.mobility.initFromDisplayString = false
```

Appendix 25 – Macro Cells Floorplan Parameters

```
############### CA configuration ##################
*.carrierAggregation.numComponentCarriers = 1
*.carrierAggregation.componentCarrier[0].numerologyIndex = ${u=1}
*.carrierAggregation.componentCarrier[0].carrierFrequency = 3.5GHz
*.carrierAggregation.componentCarrier[0].numBands = ${numRbs=273}

*.gnb*.cellularNic.numCarriers = 1
*.gnb*.cellularNic.channelModel[0].componentCarrierIndex = 0
*.ue[*].cellularNic.numCarriers = 1
*.ue[*].cellularNic.nrChannelModel[0].componentCarrierIndex = 0
```

Appendix 26 – Macro Cells CA Configurations

```
################ BS position ##################
*.gnb1.mobility.initialX = 680m
*.gnb1.mobility.initialY = 680m
*.gnb2.mobility.initialX = 2040m
*.gnb2.mobility.initialY = 680m
```

Appendix 27 – Macro Cells BS Parameters

```
############### UE configuration ###################
*.numUe = ${numUEs=140}
```

Appendix 28 – Macro Cells Number of UE

```
############# UEs position ##################
*.ue[*].mobility.typename = "LinearMobility"
*.ue[*].mobility.initialX = uniform(0m,2720m)
*.ue[*].mobility.initialY = uniform(0m,1360m)
*.ue[*].mobility.speed = uniform(3mps,13.89mps)
```

Appendix 29 – Macro Cells UE Position

```
#============= Application Setup =============
*.ue[*].app[0].typename = "CbrReceiver"
*.ue[*].app[0].localPort = 3000

*.CBR_Server.app[*].typename = "CbrSender"
*.CBR_Server.app[*].PacketSize = 1000
*.CBR_Server.app[*].sampling_time = 0.02s
*.CBR_Server.app[*].destAddress = "ue[" + string(ancestorIndex(0)) + "]"
*.CBR_Server.app[*].destPort = 3000
*.CBR_Server.app[*].localPort = 3088+ancestorIndex(0)
*.CBR_Server.app[*].startTime = uniform(0s,0.02s)
*.CBR_Server.app[*].finishTime = uniform(1199.9s,1199.92s)
```

Appendix 30 - Macro Cells CBR Parameters

```
##============= Application Setup =============
*.ue[*].app[1].typename = "VoIPReceiver"
*.ue[*].app[1].localPort = 4000

*.VoIP_Server.app[*].PacketSize = 100
*.VoIP_Server.app[*].destAddress = "ue[" + string(ancestorIndex(0)) + "]"
*.VoIP_Server.app[*].destPort = 4000
*.VoIP_Server.app[*].localPort = 4088+ancestorIndex(0)
*.VoIP_Server.app[*].typename = "VoIPSender"
*.VoIP_Server.app[*].startTime = uniform(0s,0.02s)
*.VoIP_Server.app[*].finishTime = uniform(1199.9s,1199.92s)
```

Appendix 31 - Macro Cells VoIP Parameters